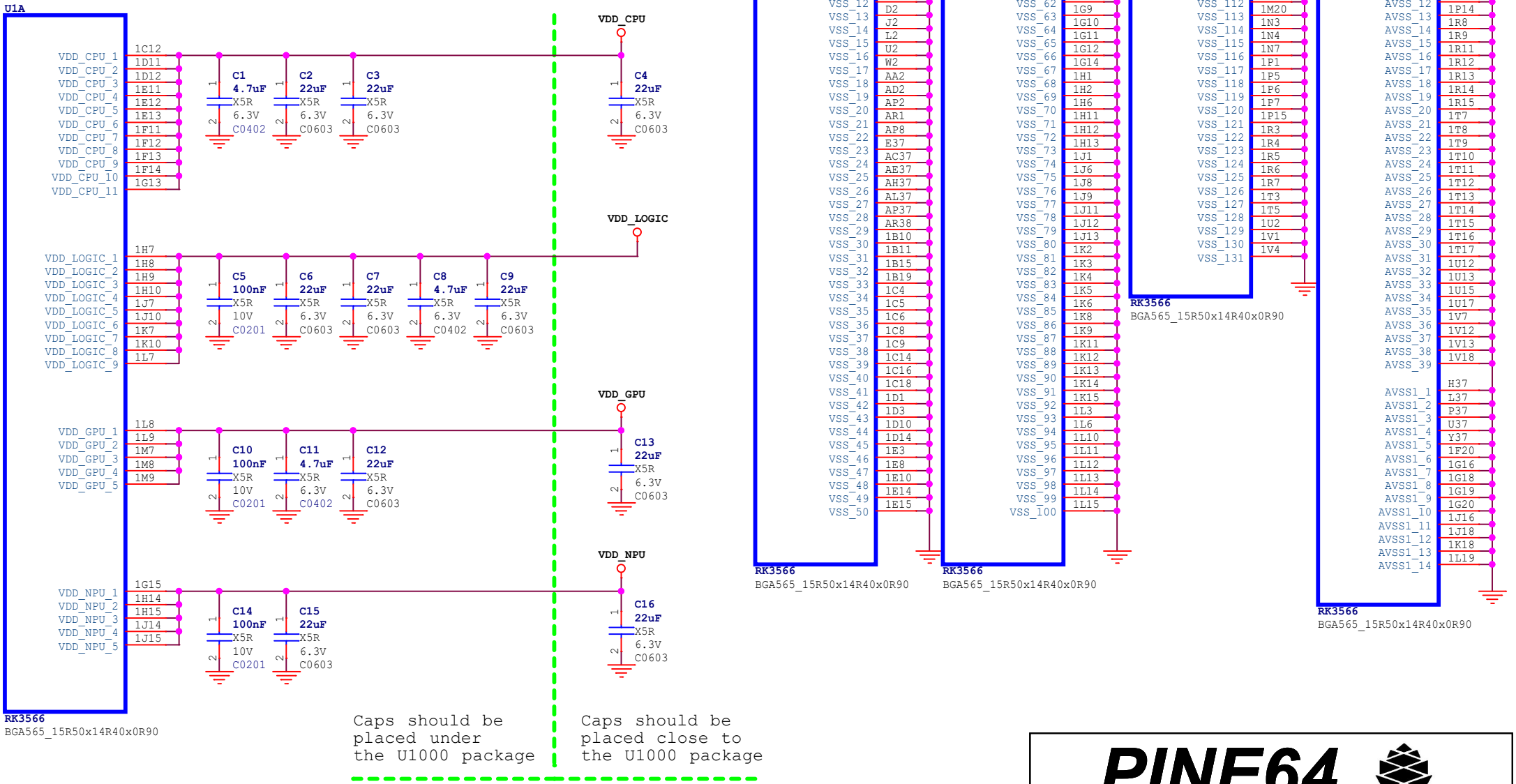
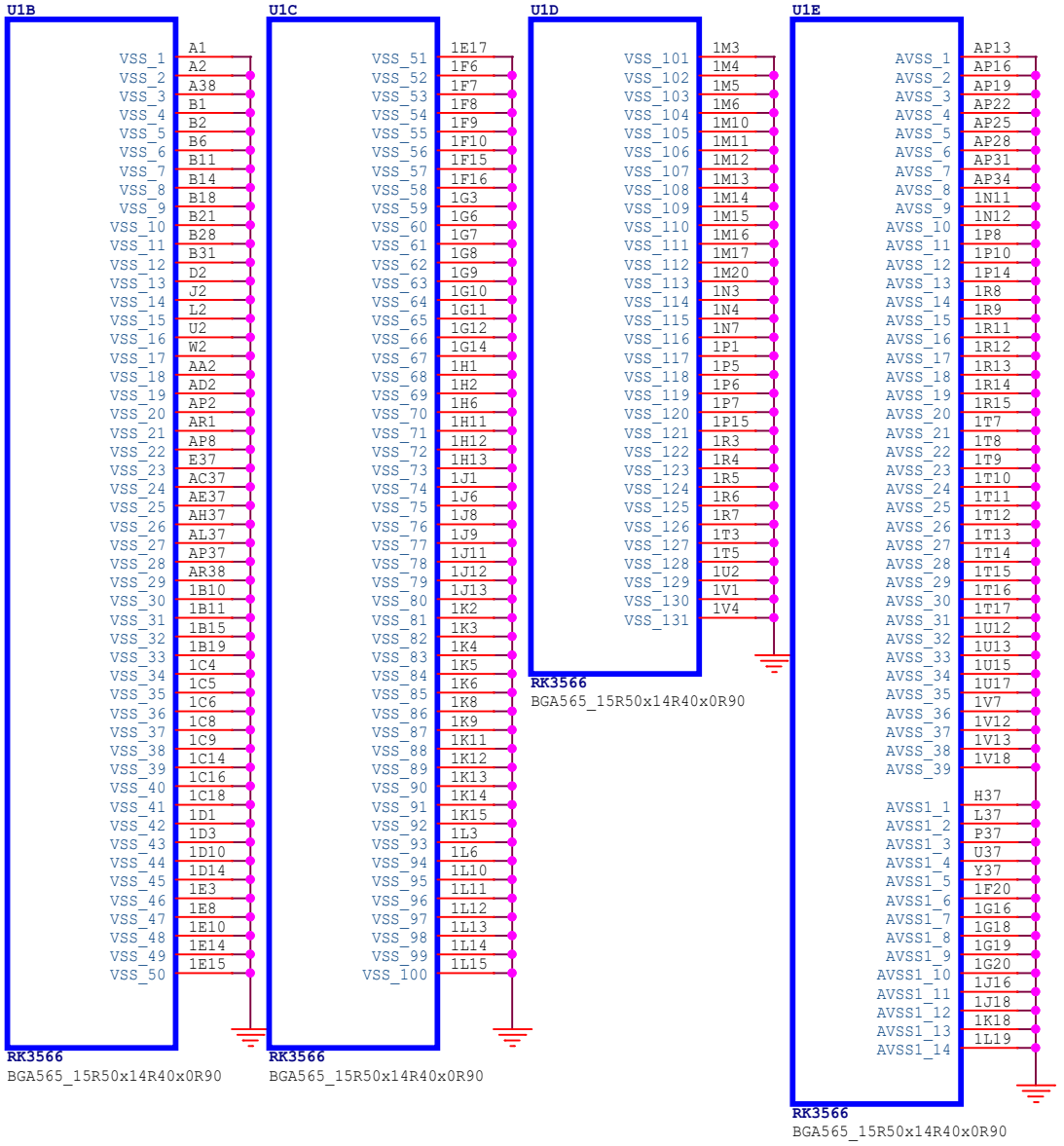
RK3566_ABCDE

(Power&GND)

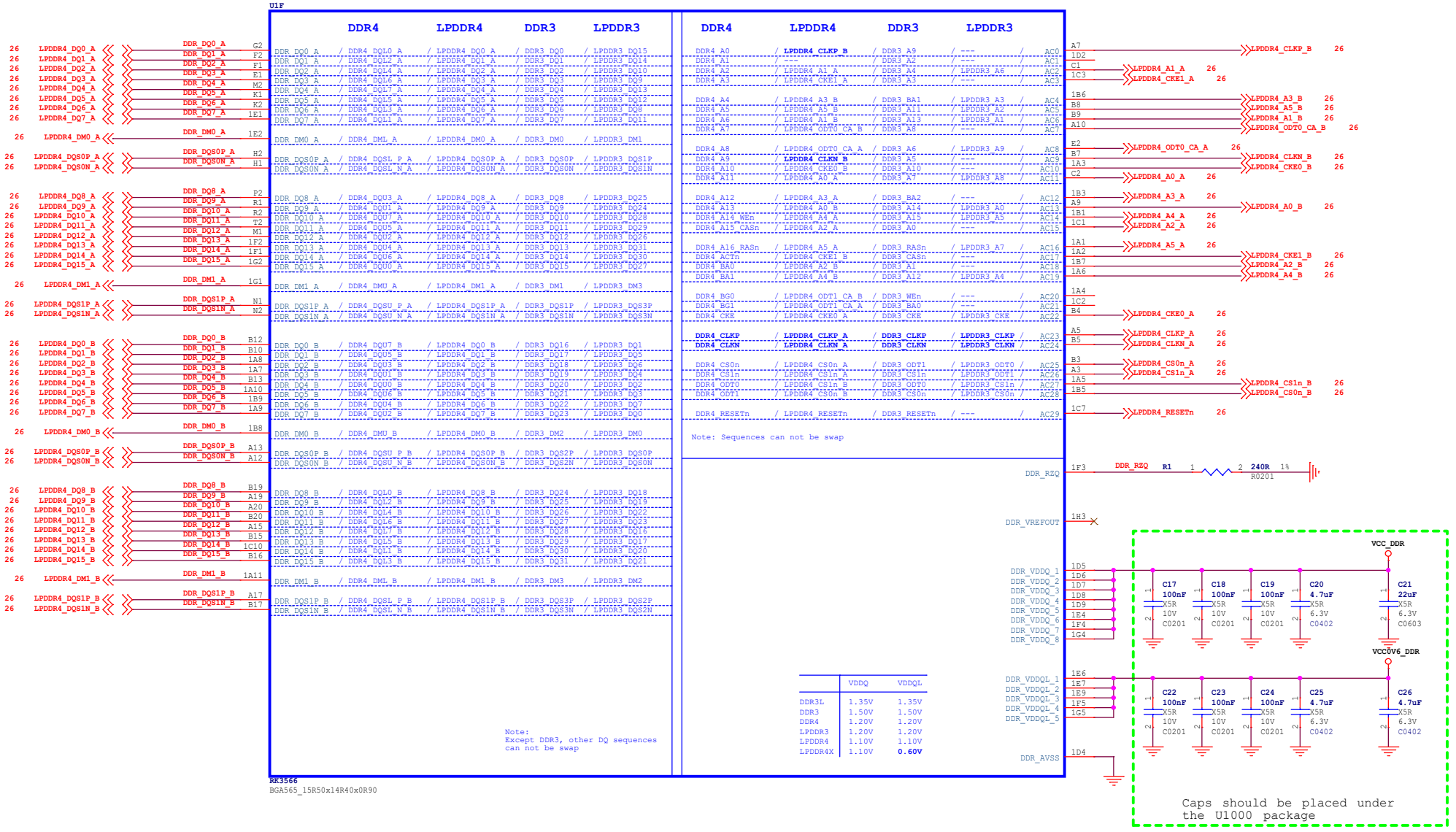




PINE64

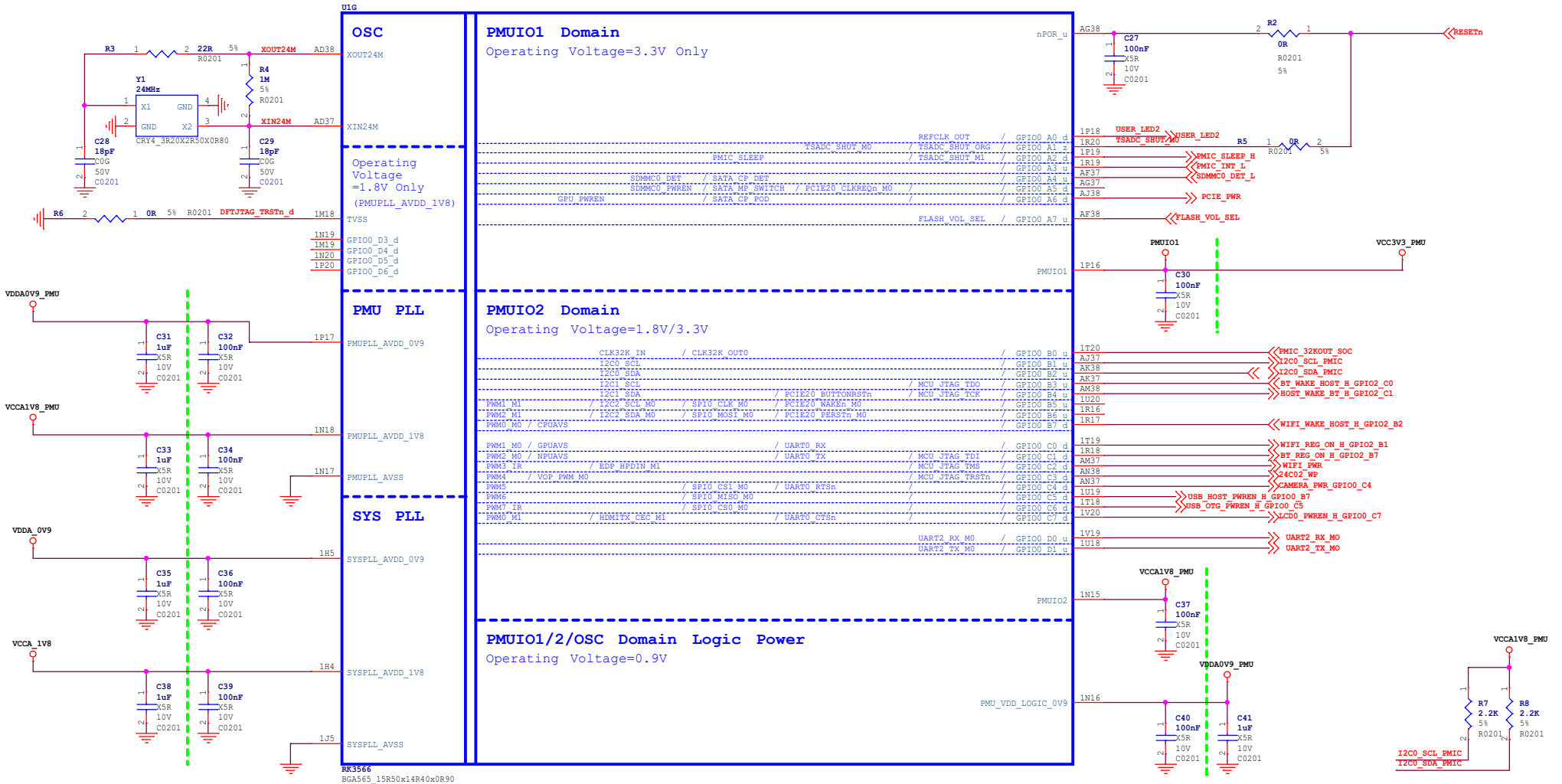
Size	Title: Quartz64 Model-B Schematic-20220124	REV
A4	Page Name: RK3566_Power/GND	V1.3
Date: Monday, January 24, 2022	Sheet 01 of 26	

RK3566_F(DDR PHY)



Size	Title:	Quartz64 Model-B Schematic-20220124	REV
A3	Page Name:	RK3566_DDR PHY	V1.3
Date:	Monday, January 24, 2022	Sheet 02 of 26	

RK3566_G(OSC/PLL/PMUIO1/2)

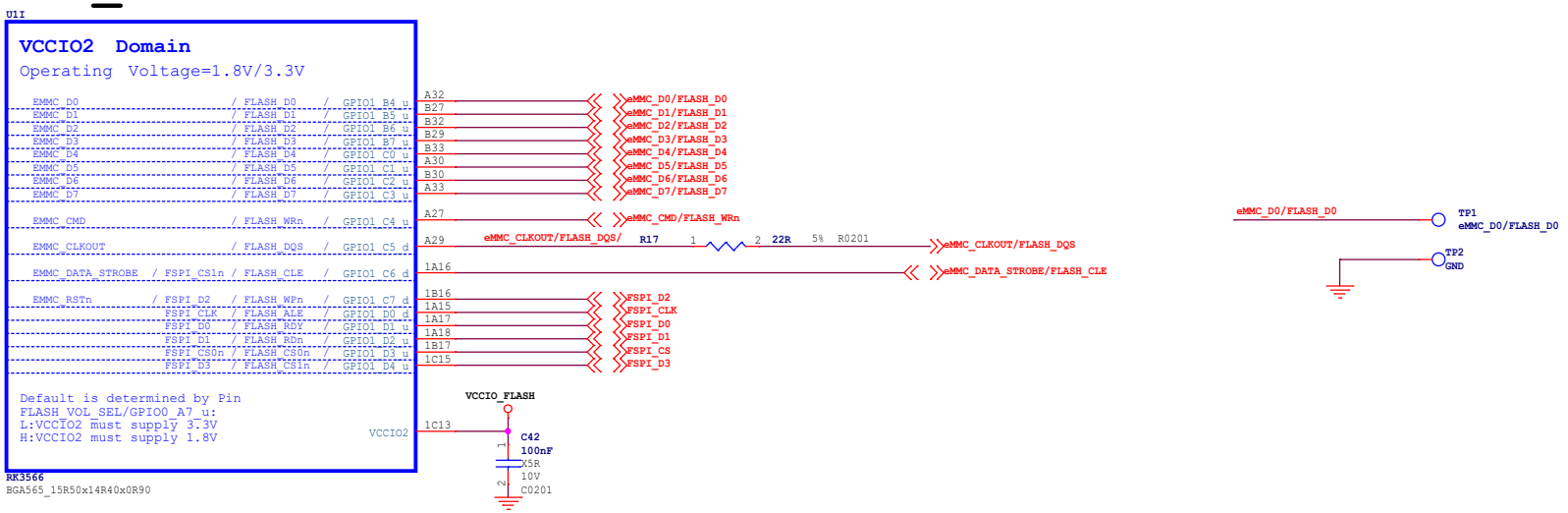


Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

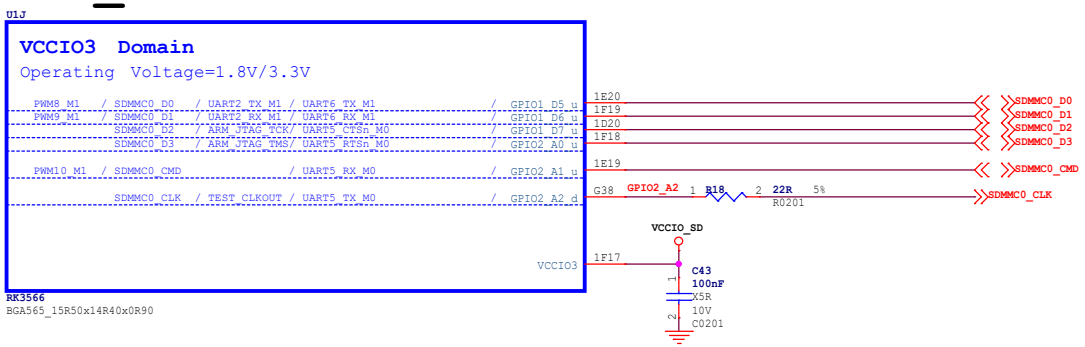
PINE64

Size	Title: Quartz64 Model-B Schematic-20220124	REV
A3	Page Name: RK3566_OSC/PLL/PMUIO	V1.3
Date: Monday, January 24, 2022	Sheet 03 of 26	

RK3566_I(VCCIO2 Domain)

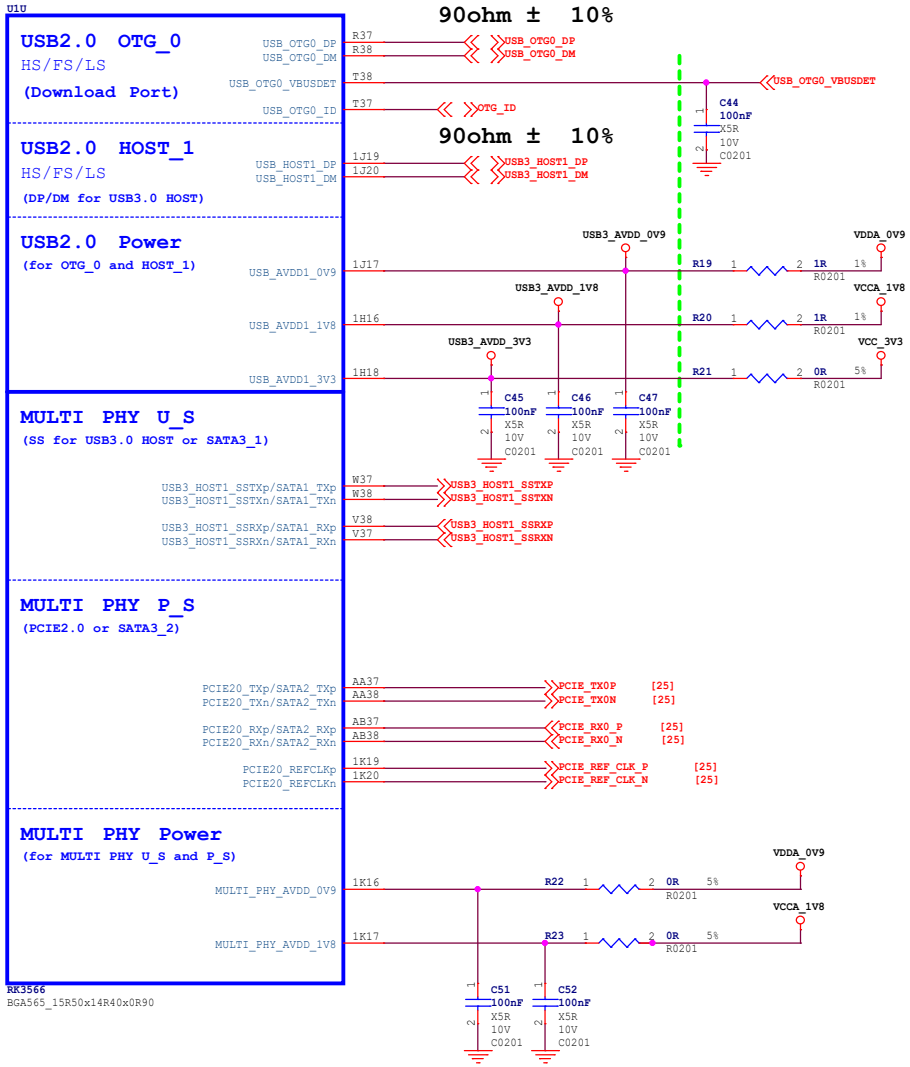


RK3566_J(VCCIO3 Domain)



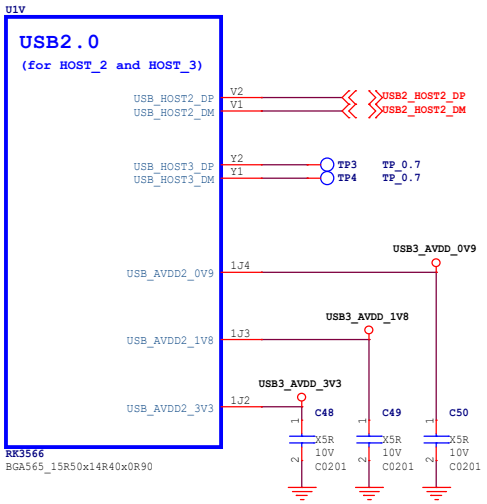
PINE64		
Size A3	Title: Quartz64 Model-B Schematic-20220124	REV V1.3
	Page Name: RK3566_Flash/SD Controller	
Date: Monday, January 24, 2022	Sheet 04 of 26	

RK3566_U(USB3.0/SATA/QSGMII/PCIe2.0 x1)



Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

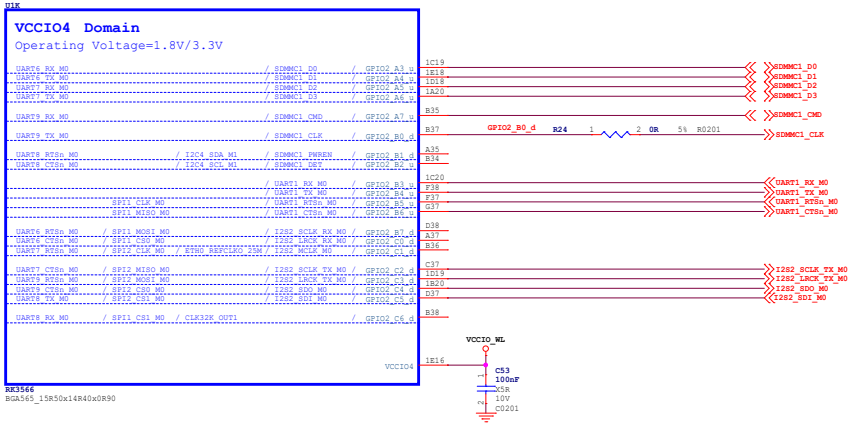
RK3566_V(USB2.0 HOST)



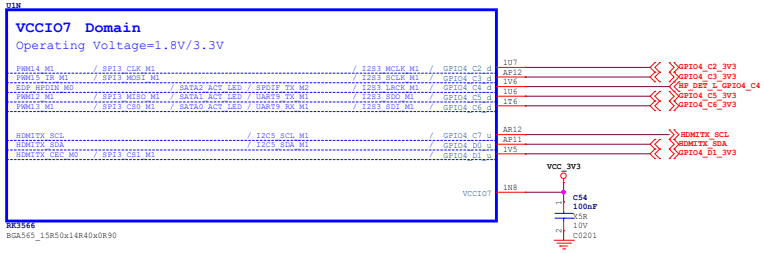
PINE64

Size A3	Title: Quartz64 Model-B Schematic-20220124	REV V1.3
	Page Name: RK3566_USB/PCIe/SATA PHY	
Date: Monday, January 24, 2022	Sheet 05 of 26	

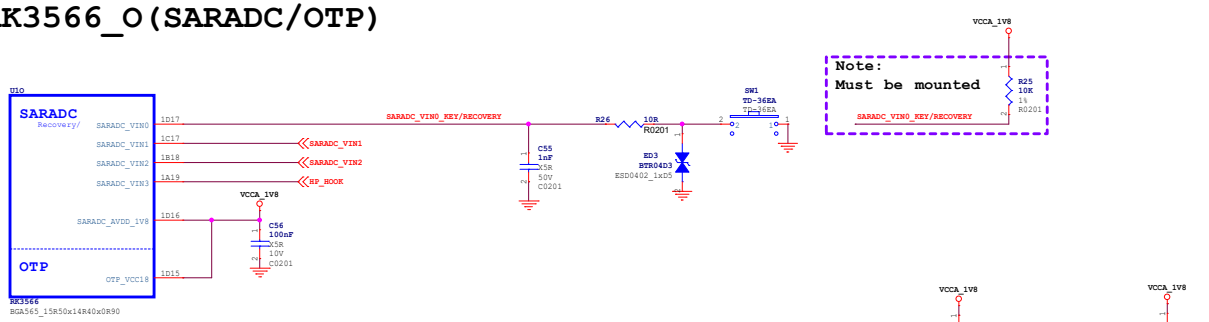
RK3566_K(VCCIO4 Domain)

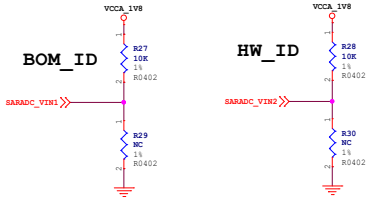


RK3566_N(VCCIO7 Domain)



RK3566_O(SARADC/OTP)

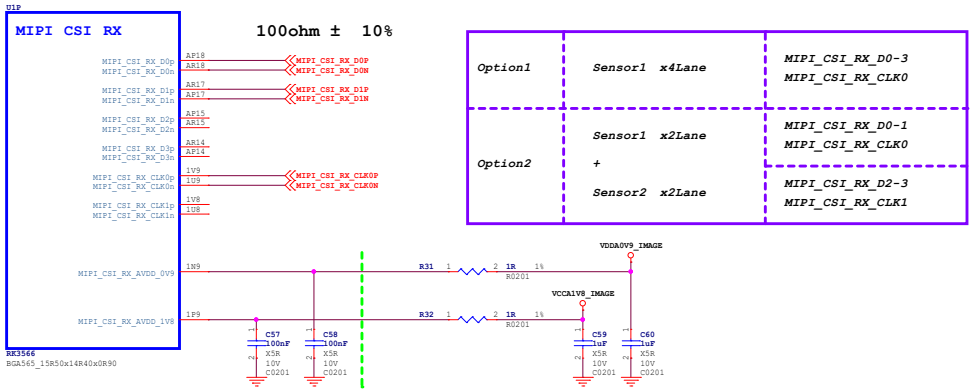




PINE64

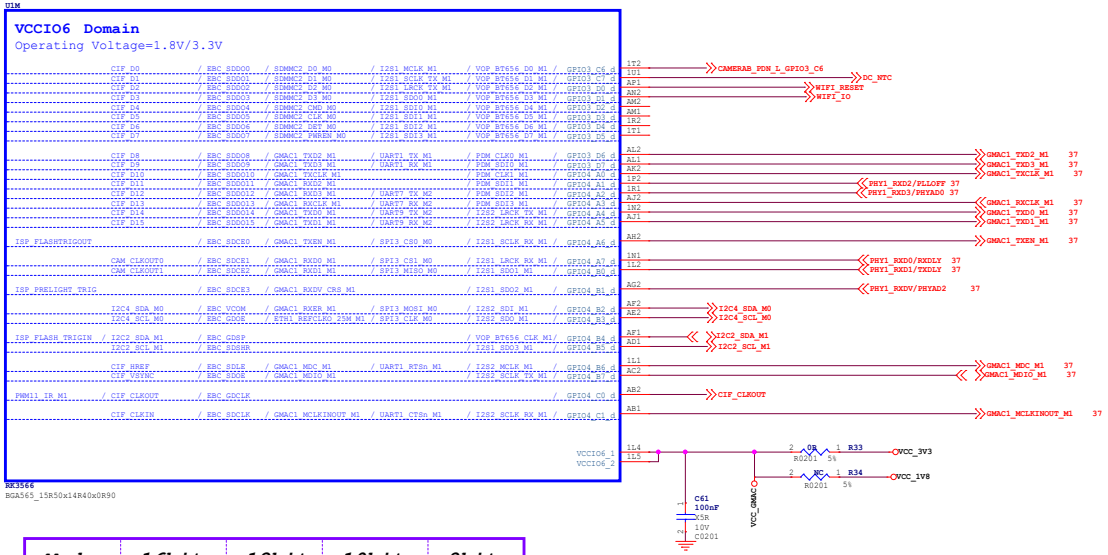
Size	Title:	REV
A2	Quartz64 Model-B Schematic-20220124	V1.3
	Page Name: RK3566_SARADC/GPIO	
Date: Monday, January 24, 2022	Sheet 06 of 26	

RK3566_P(MIPI_CSI_RX)



Option1	Sensor1 x4Lane	MIPI_CSI_RX_D0-3 MIPI_CSI_RX_CLK0
Option2	Sensor1 x2Lane + Sensor2 x2Lane	MIPI_CSI_RX_D0-1 MIPI_CSI_RX_CLK0 MIPI_CSI_RX_D2-3 MIPI_CSI_RX_CLK1

RK3566_M(VCCIO6 Domain)



Mode	16bit	12bit	10bit	8bit
CIF_D0	D0	--	--	--
CIF_D1	D1	--	--	--
CIF_D2	D2	--	--	--
CIF_D3	D3	--	--	--
CIF_D4	D4	D0	--	--
CIF_D5	D5	D1	--	--
CIF_D6	D6	D2	D0	--
CIF_D7	D7	D3	D1	--
CIF_D8	D8	D4	D2	D0
CIF_D9	D9	D5	D3	D1
CIF_D10	D10	D6	D4	D2
CIF_D11	D11	D7	D5	D3
CIF_D12	D12	D8	D6	D4
CIF_D13	D13	D9	D7	D5
CIF_D14	D14	D10	D8	D6
CIF_D15	D15	D11	D9	D7

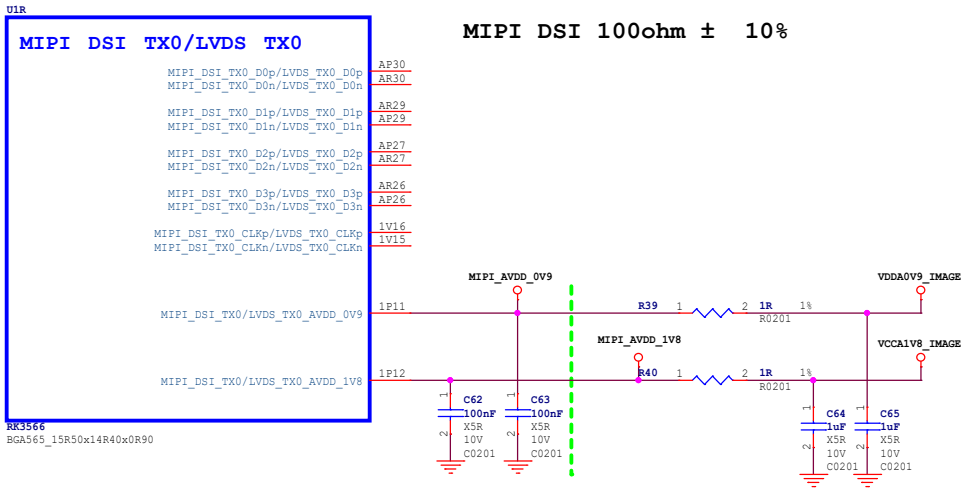
Support BT601 YCbCr 422 8bit input
Support BT656 YCbCr 422 8bit input
Support RAW 8/10/12bit input
Support BT1120 YCbCr 422 8/10/12/16bit input, single/dual-edge sampling
Support 2/4 mixed BT656/BT1120 YCbCr 422 8bit input

Note:
Caps of between dashed green lines and U1000 should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

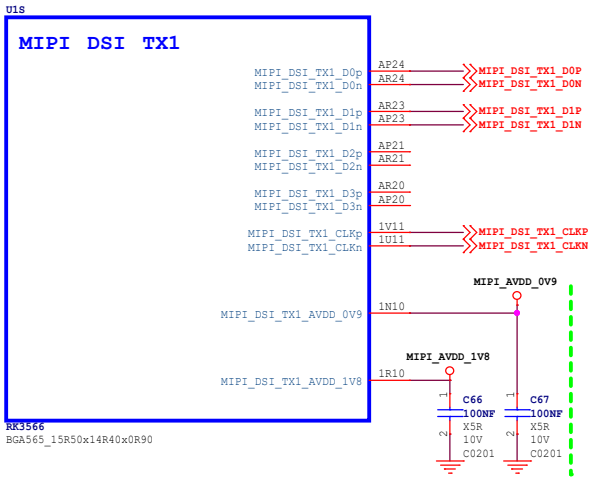
PINE64

Size: A2	Title: Quartz64 Model-B Schematic-20220124	REV: V1.3
	Page Name: RK3566_VI Interface	
Date: Monday, January 24, 2022	Sheet 07 of 26	

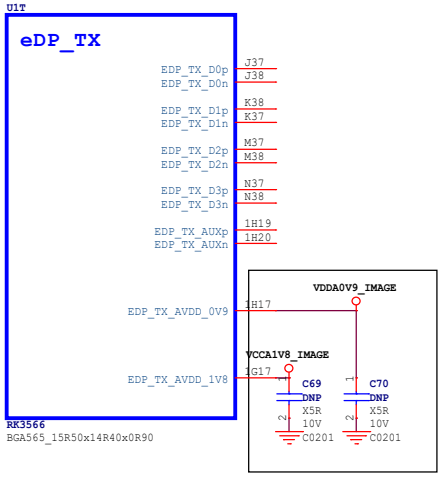
RK3566_R(MIPI_DSI_TX0/LVDS_TX0)



RK3566_S(MIPI_DSI_TX1/LVDS_TX1)



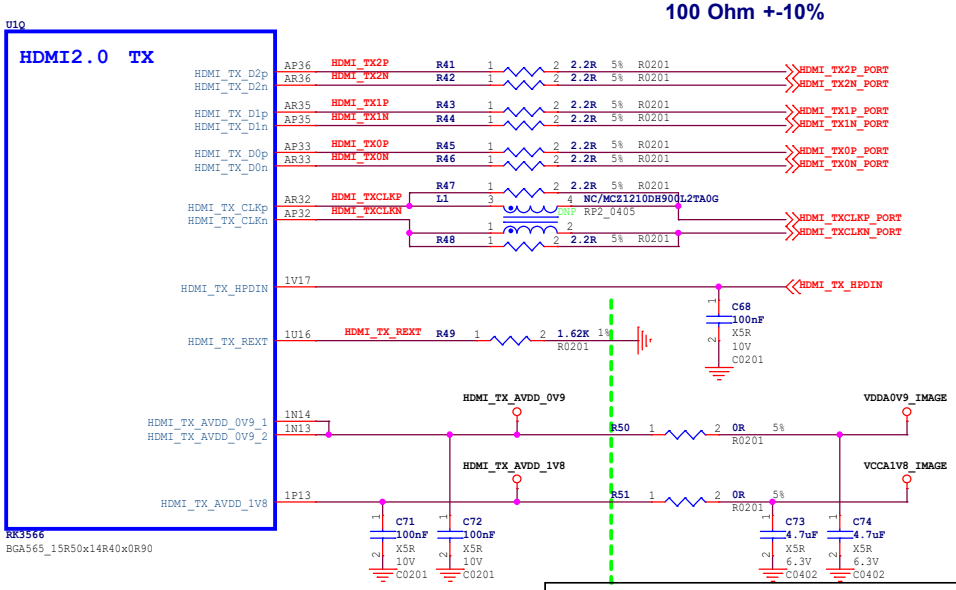
RK3566_T(eDP/DP TX)



Boxed capacitors should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

Note:
Caps of between dashed green lines and U1000
should be placed under the U1000 package.
Other caps should be placed close to the U1000 package

RK3566_Q(HDMI2.0 TX)



PINE64

Size	Title: Quartz64 Model-B Schematic-20220124	REV
A3	Page Name: RK3566_VO Interface_1	V1.3
Date: Monday, January 24, 2022	Sheet 08 of 26	

RK3566_L(VCCIO5 Domain)

U1L

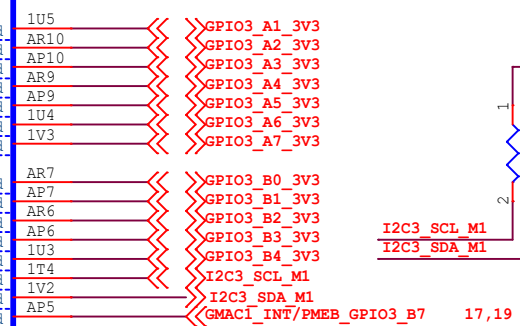
VCCIO5 Domain

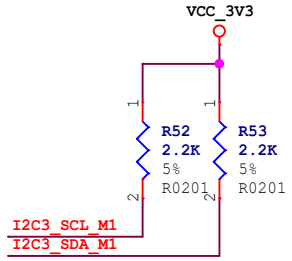
Operating Voltage=1.8V/3.3V

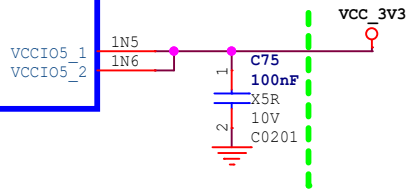
	VOP_BT1120_D0	/ SPI1_CS0_M1		/ SDMMC2_D0_M1	/ GPIO3_A1_d
	VOP_BT1120_D1		/ GMAC1_TXD2_M0 / I2S3_MCLK_M0	/ SDMMC2_D1_M1	/ GPIO3_A2_d
	VOP_BT1120_D2		/ GMAC1_TXD3_M0 / I2S3_SCLK_M0	/ SDMMC2_D2_M1	/ GPIO3_A3_d
	VOP_BT1120_D3		/ GMAC1_RXD2_M0 / I2S3_LRCK_M0	/ SDMMC2_D3_M1	/ GPIO3_A4_d
	VOP_BT1120_D4		/ GMAC1_RXD3_M0 / I2S3_SDO_M0	/ SDMMC2_CMD_M1	/ GPIO3_A5_d
	VOP_BT1120_CLK		/ GMAC1_TXCLK_M0 / I2S3_SDI_M0	/ SDMMC2_CLK_M1	/ GPIO3_A6_d
	VOP_BT1120_D5		/ GMAC1_RXCLK_M0	/ SDMMC2_DET_M1	/ GPIO3_A7_d
	VOP_BT1120_D6		/ ETH1_REFCLKO_25M_M0	/ SDMMC2_PWREN_M1	/ GPIO3_B0_d
PWM8_M0	/ VOP_BT1120_D7		/ GMAC1_RXD0_M0 / UART4_RX_M1		/ GPIO3_B1_d
PWM9_M0	/ VOP_BT1120_D8		/ GMAC1_RXD1_M0 / UART4_TX_M1		/ GPIO3_B2_d
	VOP_BT1120_D9	/ I2C5_SCL_M0	/ GMAC1_RXDV_CRS_M0	/ PDM_SDI0_M2	/ GPIO3_B3_d
	VOP_BT1120_D10	/ I2C5_SDA_M0	/ GMAC1_RXER_M0	/ PDM_SDI1_M2	/ GPIO3_B4_d
PWM10_M0	/ VOP_BT1120_D11	/ I2C3_SCL_M1	/ GMAC1_TXD0_M0		/ GPIO3_B5_d
PWM11_IR_M0	/ VOP_BT1120_D12	/ I2C3_SDA_M1	/ GMAC1_TXD1_M0		/ GPIO3_B6_d
PWM12_M0			/ GMAC1_TXEN_M0 / UART3_TX_M1	/ PDM_SDI2_M2	/ GPIO3_B7_d
PWM13_M0			/ GMAC1_MCLKINOUT_M0 / UART3_RX_M1	/ PDM_SDI3_M2	/ GPIO3_C0_d
	VOP_BT1120_D13	/ SPI1_MOSI_M1	/ PCIE20_PERSTn_M1	/ I2S1_SDO2_M2	/ GPIO3_C1_d
	VOP_BT1120_D14	/ SPI1_MISO_M1	/ UART5_TX_M1	/ I2S1_SDO3_M2	/ GPIO3_C2_d
	VOP_BT1120_D15	/ SPI1_CLK_M1	/ UART5_RX_M1	/ I2S1_SCLK_RX_M2	/ GPIO3_C3_d
PWM14_M0		/ VOP_PWM_M1	/ GMAC1_MDC_M0 / UART7_TX_M1	/ PDM_CLK1_M2	/ GPIO3_C4_d
PWM15_IR_M0		/ SPDIF_TX_M1	/ GMAC1_MDIO_M0 / UART7_RX_M1	/ I2S1_LRCK_RX_M2	/ GPIO3_C5_d

RK3566

BGA565_15R50x14R40x0R90







Note:

Caps of between dashed green lines and U1000 should be placed under the U1000 package

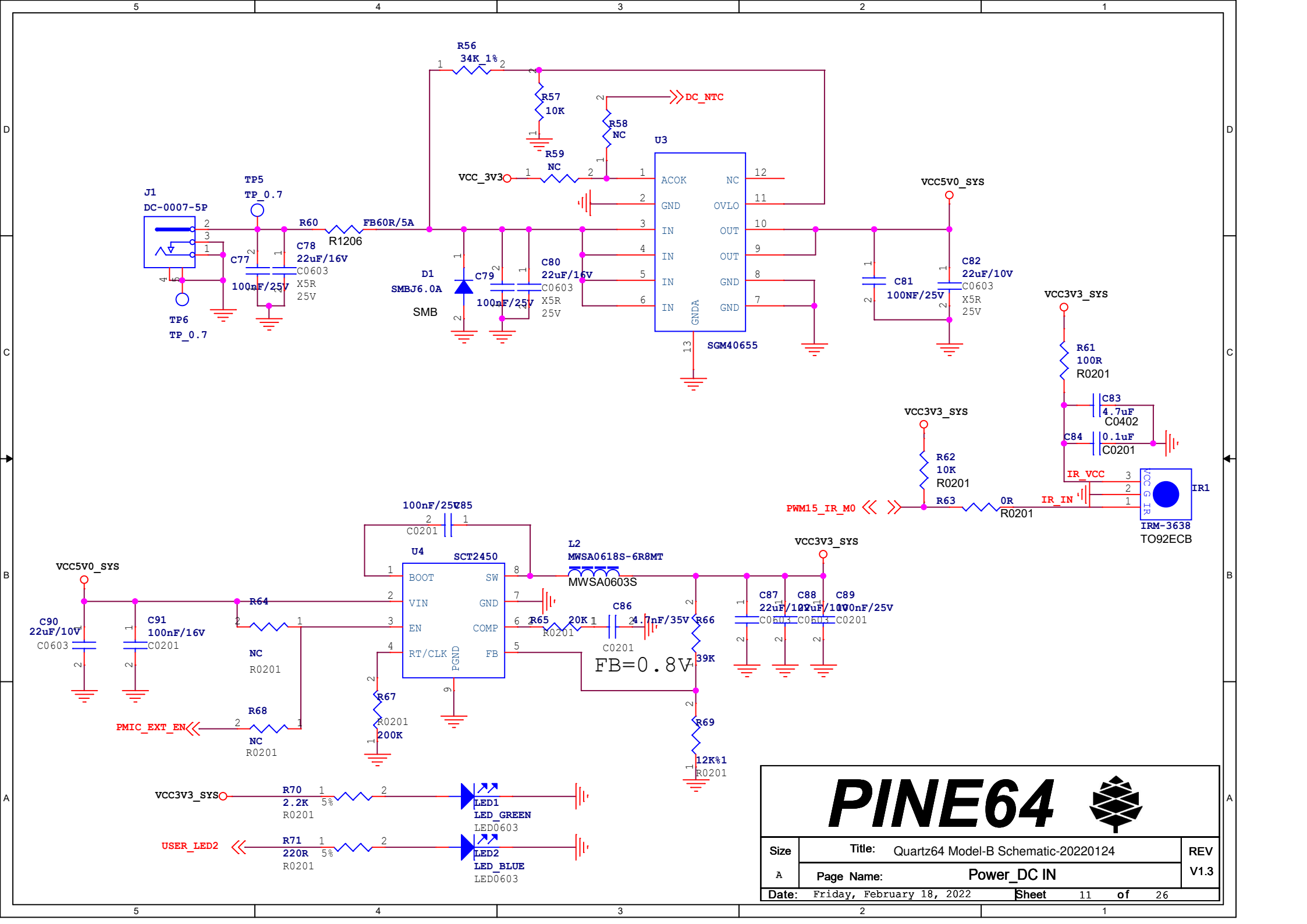
PINE64

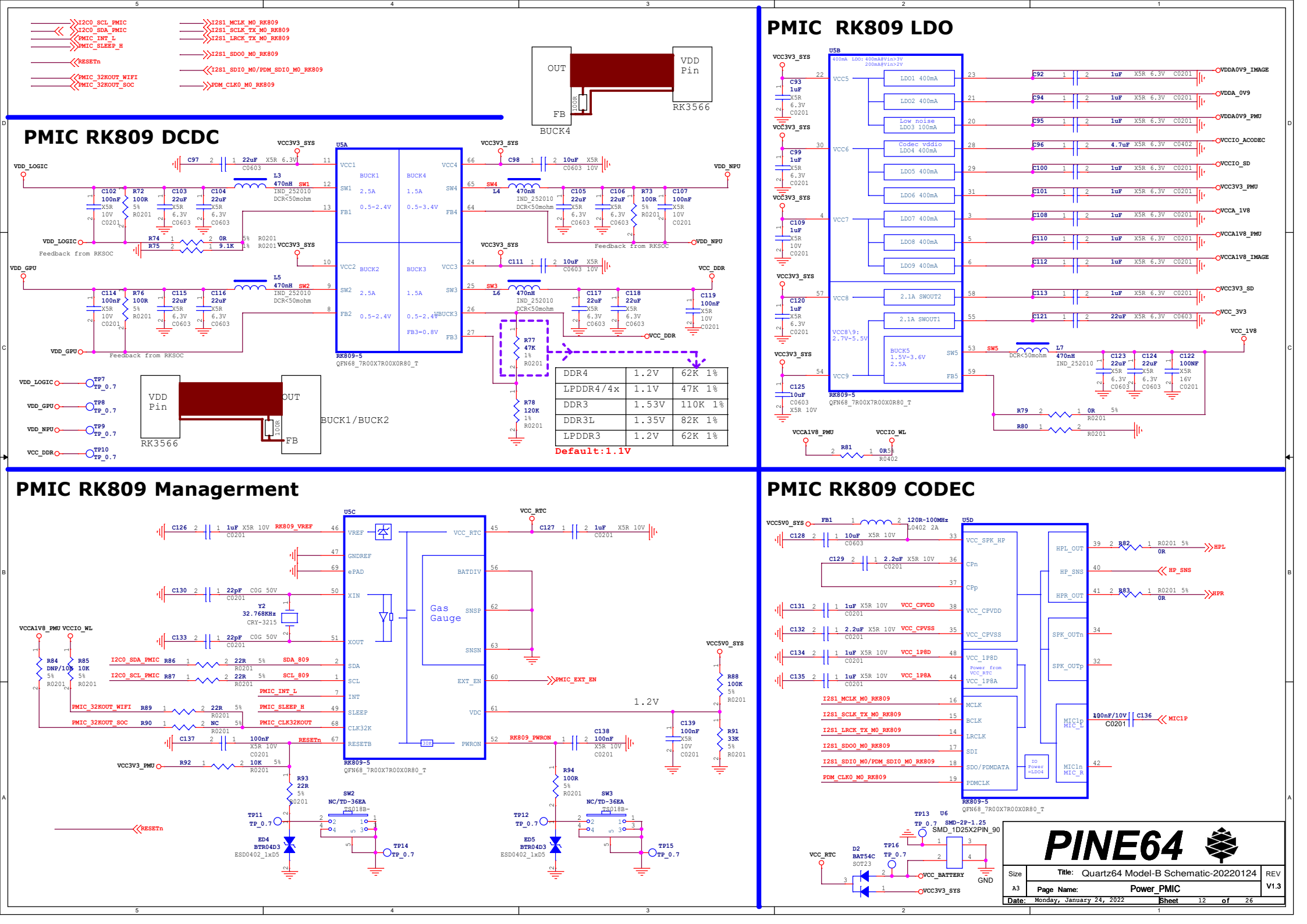
Size A4	Title: Quartz64 Model-B Schematic-20220124	REV V1.3
	Page Name: RK3566_VO Interface_2	
Date: Monday, January 24, 2022	Sheet 09 of 26	

RK3566_H(VCCIO1 Domain)



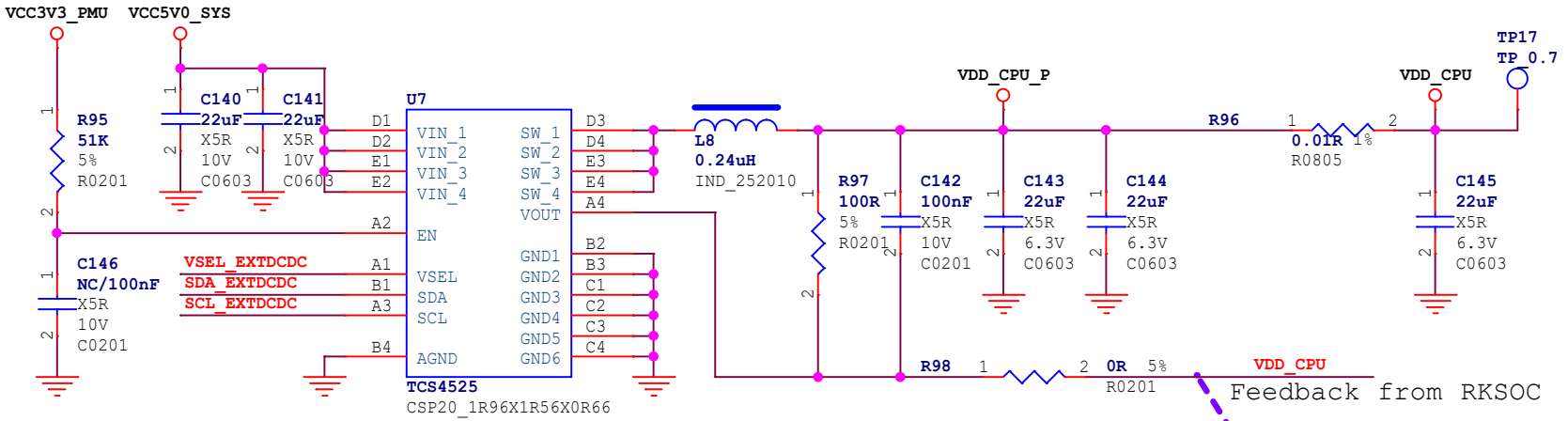
PINE64		
Size: A4	Title: Quartz64 Model-B Schematic-20220124	REV: V1.3
	Page Name: RK3566_Audio Interface	
Date: Monday, January 24, 2022	Sheet 10 of 26	

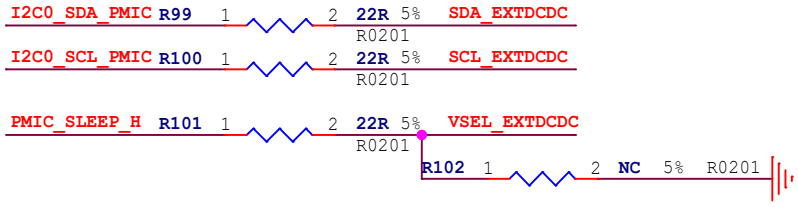

R56
34K_1%
R57
10K
R58
NC
R59
NC
DC_NTC
VCC_3V3
U3
ACOK
GND
IN
IN
IN
GNDA
NC
OVLO
OUT
OUT
GND
GND
SGM40655
J1
DC-0007-5P
TP5
TP_0.7
TP6
TP_0.7
R60
FB60R/5A
R1206
C77
100nF/25V
C78
22uF/16V
C0603
X5R
25V
D1
SMBJ6.0A
SMB
C79
100nF/25V
C80
22uF/16V
C0603
X5R
25V
VCC5V0_SYS
C81
100NF/25V
C82
22uF/10V
C0603
X5R
25V
VCC3V3_SYS
R61
100R
R0201
C83
4.7uF
C0402
C84
0.1uF
C0201
IR_VCC
IR_IN
IR1
IRM-3638
TO92ECB
PWM15_IR_M0
R62
10K
R0201
R63
0R
R0201
VCC5V0_SYS
C90
22uF/10V
C0603
C91
100nF/16V
C0201
R64
NC
R0201
R68
NC
R0201
PMIC_EXT_EN
100nF/25V
C0201
U4
SCT2450
BOOT
VIN
EN
RT/CLK
SW
GND
COMP
FB
PGND
L2
MWSA0618S-6R8MT
MWSA0603S
R65
20K
R0201
C86
4.7nF/35V
C0201
FB=0.8V
R66
39K
R67
R0201
200K
R69
12K%1
R0201
VCC3V3_SYS
C87
22uF/10V
C0603
C88
22uF/10V
C0603
C89
100nF/25V
C0201
VCC3V3_SYS
R70
2.2K
R0201
LED1
LED_GREEN
LED0603
USER_LED2
R71
220R
R0201
LED2
LED_BLUE
LED0603
PINE64
Size
A
Title: Quartz64 Model-B Schematic-20220124
Page Name: Power_DC IN
REV
V1.3
Date: Friday, February 18, 2022
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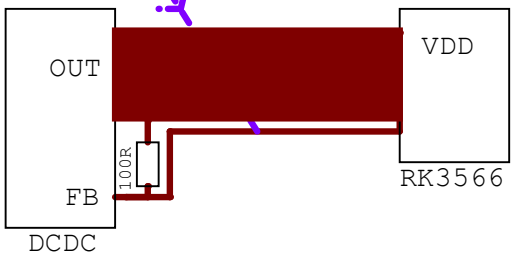

I2C0_SCL_PMIC
I2C0_SDA_PMIC
PMIC_INT_L
PMIC_SLEEP_H
RESETn
PMIC_32KOUT_WIFI
PMIC_32KOUT_SOC
I2S1_MCLK_M0_RK809
I2S1_SCLK_TX_M0_RK809
I2S1_LRCK_TX_M0_RK809
I2S1_SDO0_M0_RK809
I2S1_SDI0_M0/PDM_SDI0_M0_RK809
PDM_CLK0_M0_RK809
OUT
VDD Pin
FB
100R
RK3566
BUCK4
PMIC RK809 DCDC
VCC3V3_SYS
U5A
VDD_LOGIC
VDD_GPU
VDD_NPU
VCC_DDR
C97 22uF X5R 6.3V C0603
C98 10uF X5R C0603 10V
C102 100nF X5R 10V C0201
R72 100R 5% R0201
C103 22uF X5R 6.3V C0603
C104 22uF X5R 6.3V C0603
L3 470nH SW1 IND_252010 DCR<50mohm
L4 470nH IND_252010 DCR<50mohm
C105 22uF X5R 6.3V C0603
C106 22uF X5R 6.3V C0603
R73 100R 5% R0201
C107 100nF X5R 10V C0201
R74 0R 5% R0201
R75 9.1K 1% R0201
Feedback from RKSOC
C111 10uF X5R C0603 10V
C114 100nF X5R 10V C0201
R76 100R 5% R0201
C115 22uF X5R 6.3V C0603
C116 22uF X5R 6.3V C0603
L5 470nH SW2 IND_252010 DCR<50mohm
L6 470nH IND_252010 DCR<50mohm
C117 22uF X5R 6.3V C0603
C118 22uF X5R 6.3V C0603
C119 100nF X5R 10V C0201
R77 47K 1% R0201
R78 120K 1% R0201
BUCK1 2.5A 0.5-2.4V
BUCK2 2.5A 0.5-2.4V
BUCK3 1.5A 0.5-2.4V FB3=0.8V
BUCK4 1.5A 0.5-3.4V
RK809-5
QFN68_7R00X7R00X0R80_T
DDR4 1.2V 62K 1%
LPDDR4/4x 1.1V 47K 1%
DDR3 1.53V 110K 1%
DDR3L 1.35V 82K 1%
LPDDR3 1.2V 62K 1%
Default:1.1V
TP7 TP_0.7
TP8 TP_0.7
TP9 TP_0.7
TP10 TP_0.7
BUCK1/BUCK2
PMIC RK809 LDO
U5B
VCC3V3_SYS
VCC5 VCC6 VCC7 VCC8 VCC9
LDO1 400mA
LDO2 400mA
Low noise LDO3 100mA
Codec vddio LDO4 400mA
LDO5 400mA
LDO6 400mA
LDO7 400mA
LDO8 400mA
LDO9 400mA
2.1A SWOUT2
2.1A SWOUT1
BUCK5 1.5V~3.6V 2.5A
VCC8\9: 2.7V~5.5V
VDDA0V9_IMAGE
VDDA_0V9
VDDA0V9_PMU
VCCIO_ACODEC
VCCIO_SD
VCC3V3_PMU
VCCA_1V8
VCCA1V8_PMU
VCCA1V8_IMAGE
VCC3V3_SD
VCC_3V3
VCC_1V8
PMIC RK809 Managerment
U5C
VCC_RTC
VREF GNDREF ePAD XIN XOUT SDA SCL INT SLEEP CLK32K RESETB
VCC_RTC BATDIV SNSP SNSN EXT_EN VDC PWRON
Gas Gauge
Y2 32.768KHz CRY-3215
I2C0_SDA_PMIC I2C0_SCL_PMIC
SDA_809 SCL_809
PMIC_INT_L PMIC_SLEEP_H
PMIC_32KOUT_WIFI PMIC_32KOUT_SOC
PMIC_CLK32KOUT
PMIC_EXT_EN
RK809_PWRON
RESETn
VCCA1V8_PMU VCCIO_WL
VCC3V3_PMU
VCC5V0_SYS
1.2V
SW2 NC/TD-36EA
SW3 NC/TD-36EA
ED4 BTR04D3 ESD0402_1xD5
ED5 BTR04D3 ESD0402_1xD5
PMIC RK809 CODEC
VCC5V0_SYS
U5D
VCC_SPK_HP CPn CPp VCC_CPVDD VCC_CPVSS VCC_1P8D VCC_1P8A
MCLK BCLK LRCLK SDI SDO/PDMDATA PDMCLK
HPL_OUT HP_SNS HPR_OUT SPK_OUTn SPK_OUTp MIC1p MIC_L MIC1n MIC_R
I2S1_MCLK_M0_RK809
I2S1_SCLK_TX_M0_RK809
I2S1_LRCK_TX_M0_RK809
I2S1_SDO0_M0_RK809
I2S1_SDI0_M0/PDM_SDI0_M0_RK809
PDM_CLK0_M0_RK809
HPL
HP_SNS
HPR
MIC1P
MIC1N
VCC_RTC
VCC_BATTERY
VCC3V3_SYS
GND
PINE64
Title: Quartz64 Model-B Schematic-20220124
Size A3
REV V1.3
Page Name: Power_PMIC
Date: Monday, January 24, 2022
Sheet 12 of 26

I2C0_SCL_PMIC
I2C0_SDA_PMIC
PMIC_SLEEP_H

VDD_CPU_EXT





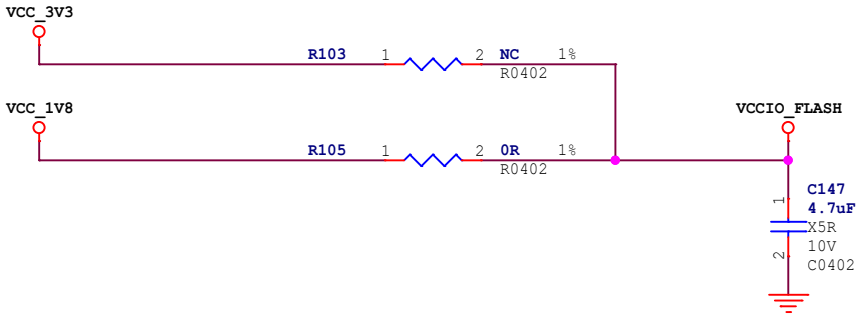


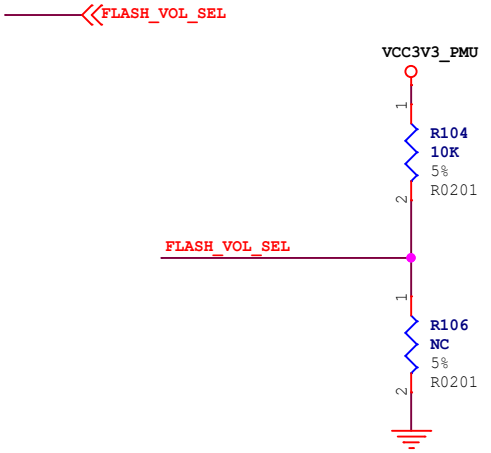
PINE64

Size: A	Title: Quartz64 Model-B Schematic-20220124	REV: V1.3
	Page Name: Power_other	
Date: Monday, January 24, 2022	Sheet 13 of 26	

Flash Power Manage

	VCCIO2 domain voltage: Recommend voltage value (VCCIO_FLASH)	FLASH_VOL_SEL state decided to VCCIO2 domain IO driven by default
eMMC	1.8V	FLASH_VOL_SEL --> Logic=H
Nand flash	Default 3.3V, Adjust according to demand 1.8V	FLASH_VOL_SEL --> Logic=L(Default)
SPI flash	Default 3.3V, Adjust according to demand 1.8V	FLASH_VOL_SEL --> Logic=L(Default)

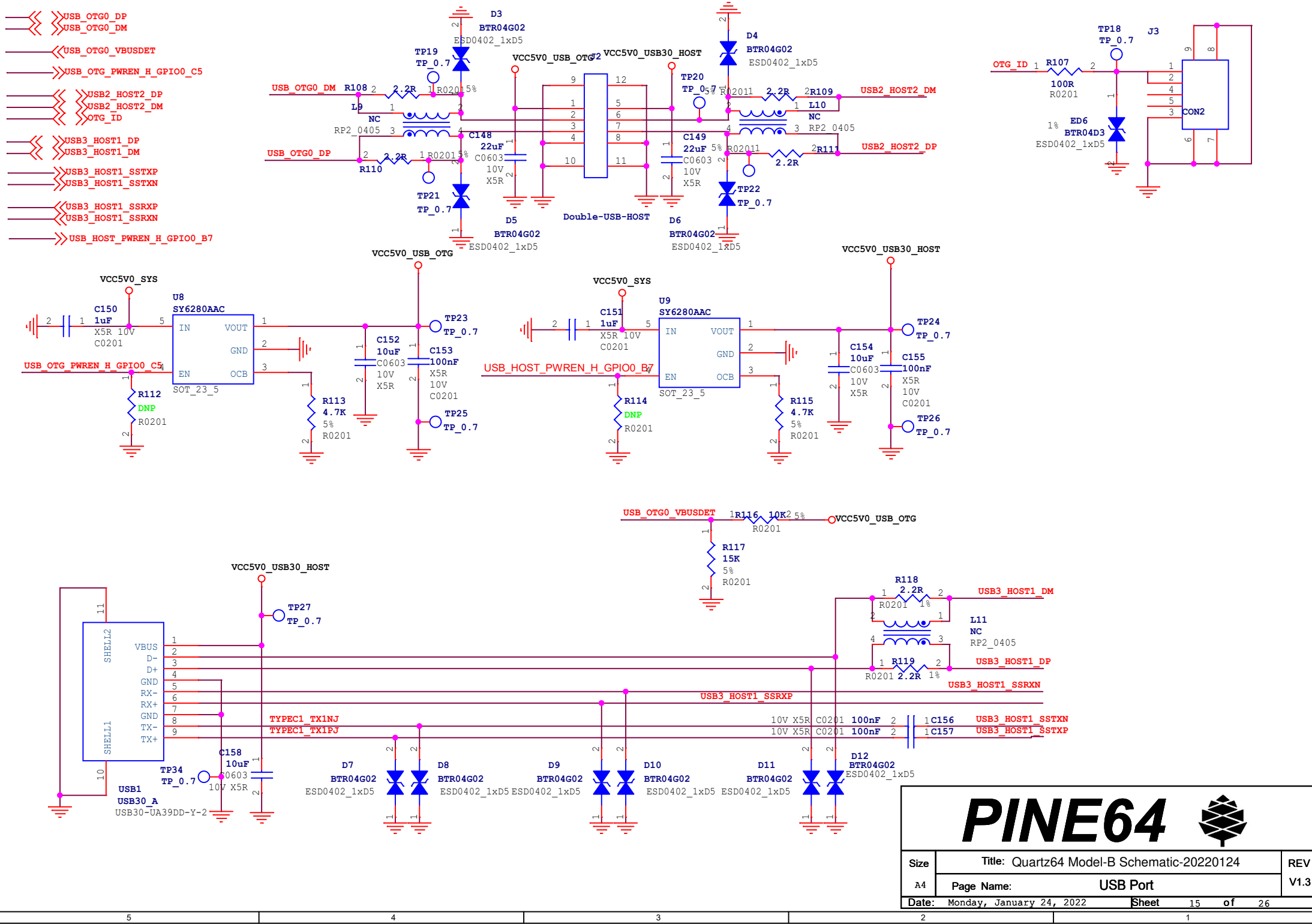




Note:
FLASH_VOL_SEL state decided
to VCCIO2 domain IO driven by default
Logic=L:3.3V IO driven
Logic=H:1.8V IO driven

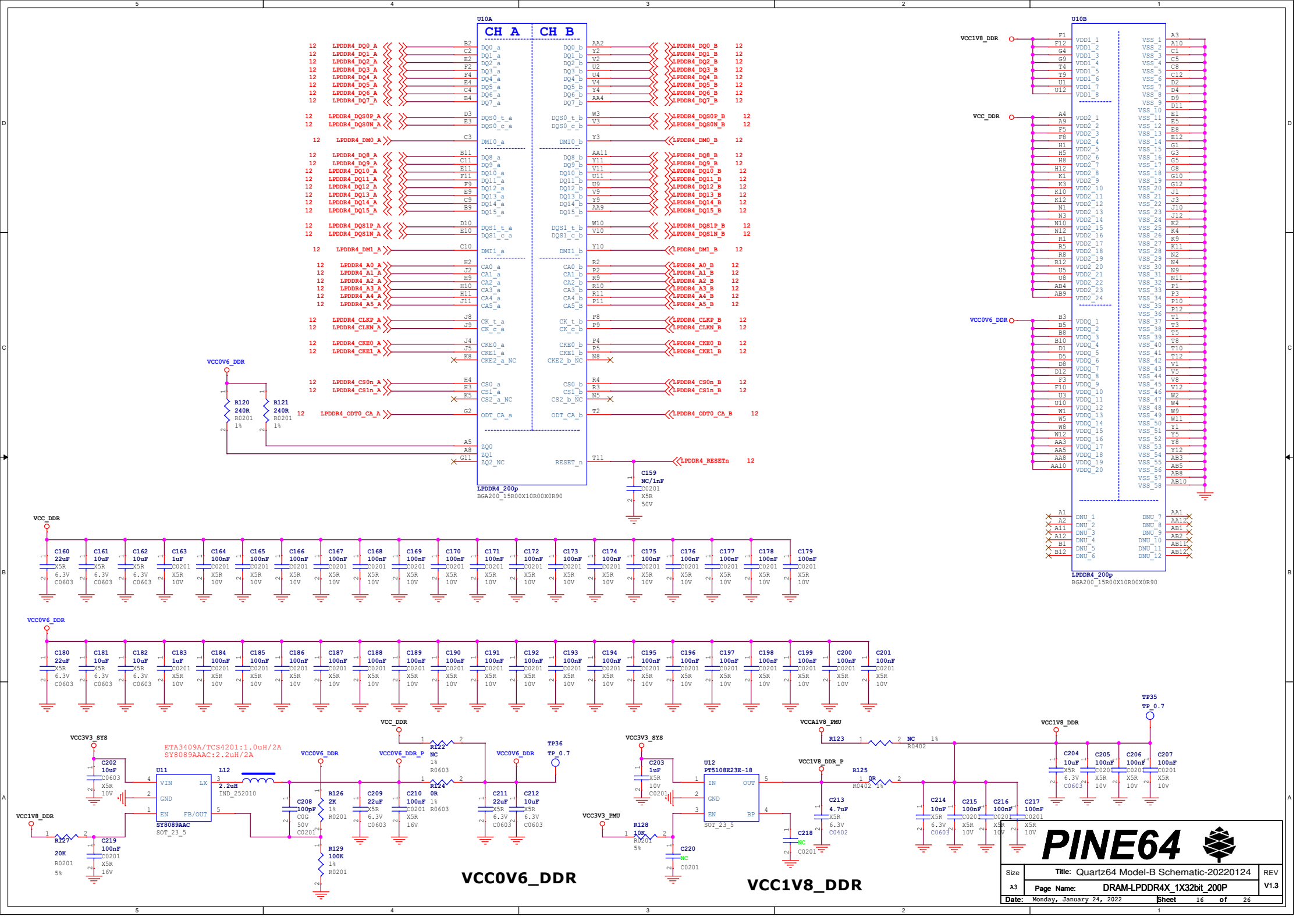
PINE64

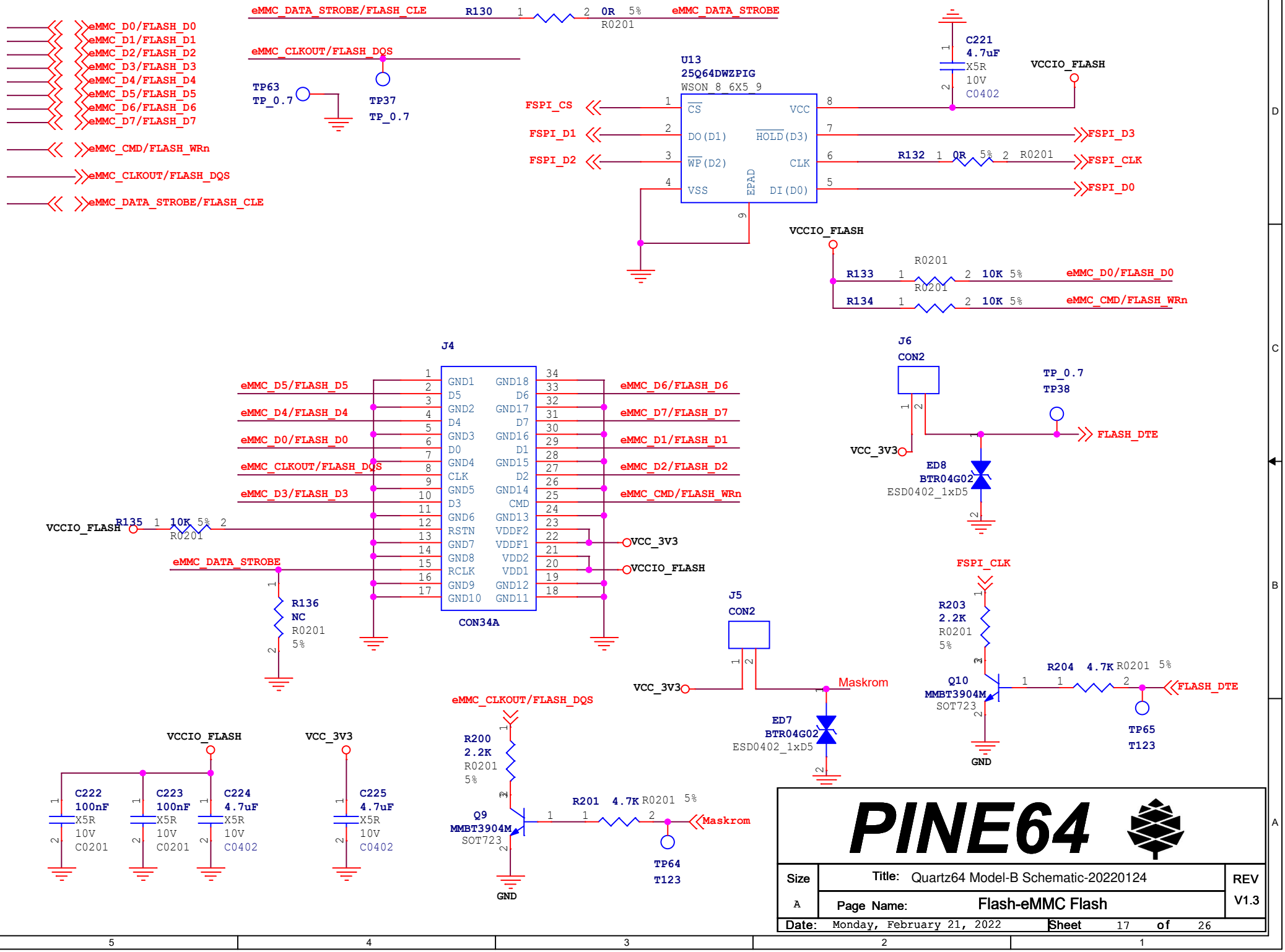
Size	Title: Quartz64 Model-B Schematic-20220124	REV
A4	Page Name: Power_Flash Power Manage	V1.3
Date: Monday, January 24, 2022	Sheet 14 of 26	

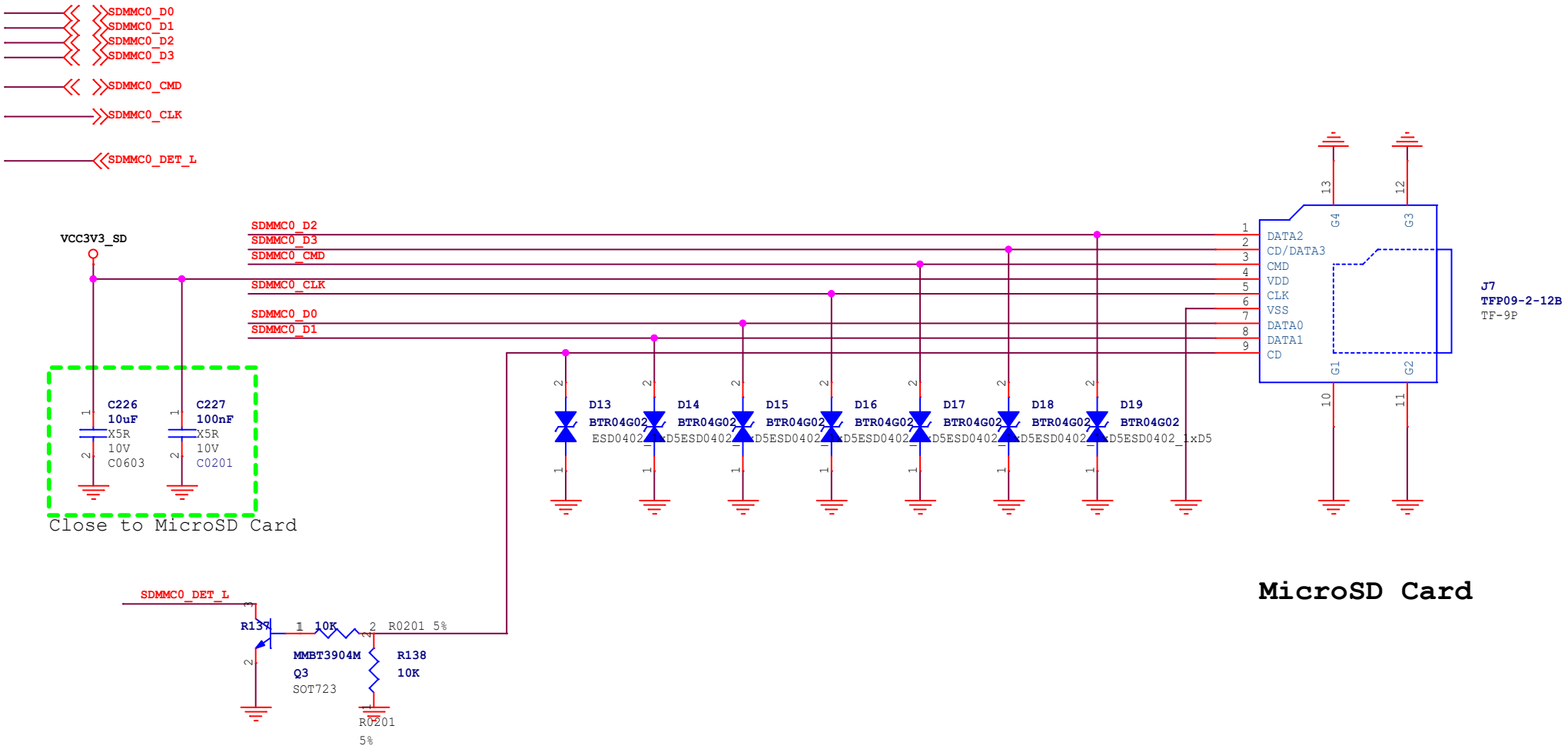


PINE64

Size: A4	Title: Quartz64 Model-B Schematic-20220124	REV: V1.3
	Page Name: USB Port	
Date: Monday, January 24, 2022	Sheet 15 of 26	

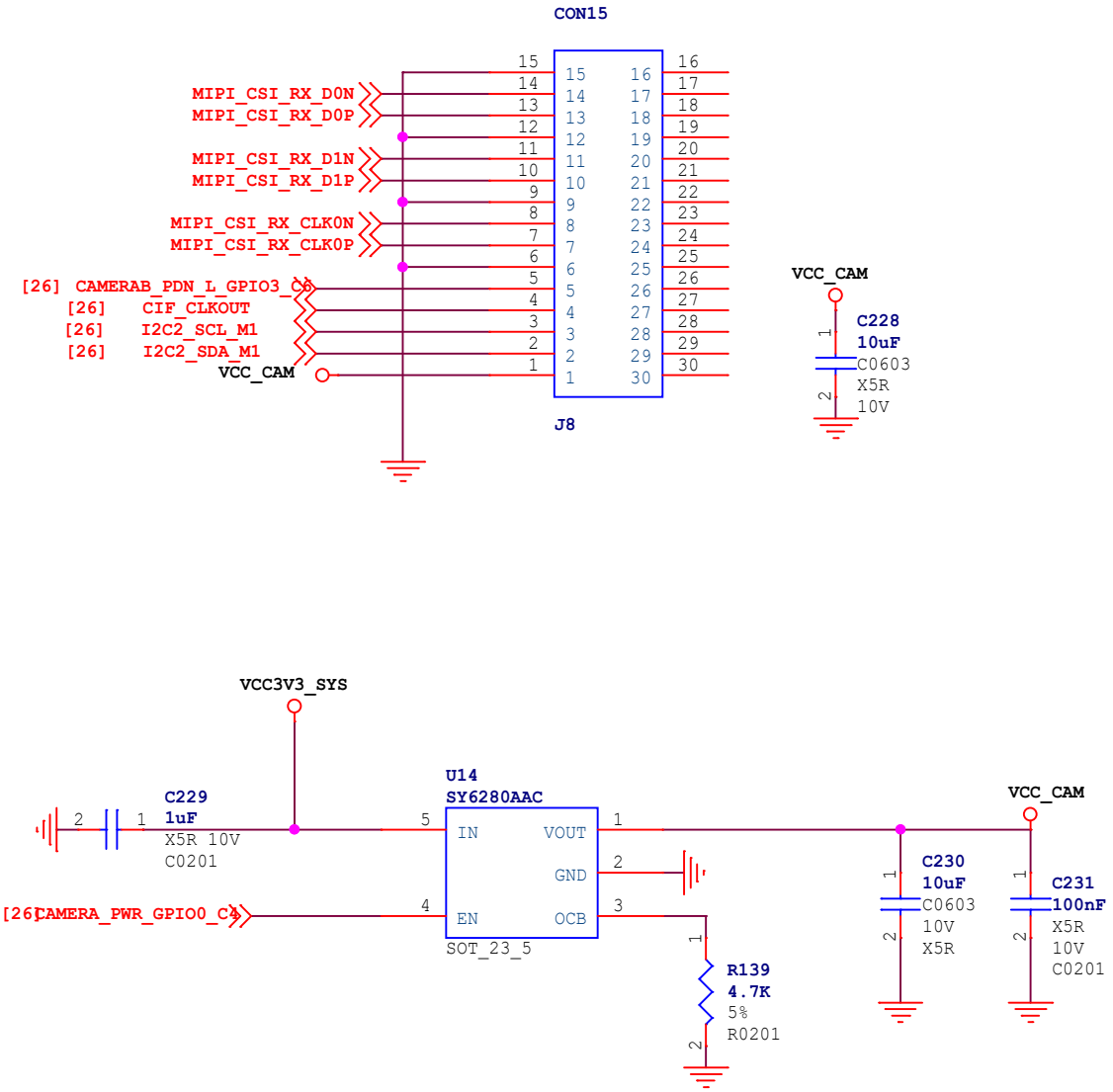

U10A
CH A
CH B
LPDDR4_200p
BGA200_15R00X10R00X0R90
U10B
LPDDR4_200p
BGA200_15R00X10R00X0R90
VCC0V6_DDR
VCC_DDR
VCC1V8_DDR
VCC3V3_SYS
VCC3V3_PMU
VCCA1V8_PMU
VCC1V8_DDR_P
VCC0V6_DDR_P
R120 240R R0201 1%
R121 240R R0201 1%
C159 NC/1nF C0201 X5R 50V
U11 SY8089AAC SOT_23_5
L12 2.2uH IND_252010
ETA3409A/TCS4201:1.0uH/2A
SY8089AAAC:2.2uH/2A
U12 PT5108E23E-18 SOT_23_5
TP35 TP_0.7
TP36 TP_0.7
VCC0V6_DDR
VCC1V8_DDR
PINE64
Title: Quartz64 Model-B Schematic-20220124
Page Name: DRAM-LPDDR4X_1X32bit_200P
Date: Monday, January 24, 2022
Sheet 16 of 26
REV V1.3
Size A3


eMMC_D0/FLASH_D0
eMMC_D1/FLASH_D1
eMMC_D2/FLASH_D2
eMMC_D3/FLASH_D3
eMMC_D4/FLASH_D4
eMMC_D5/FLASH_D5
eMMC_D6/FLASH_D6
eMMC_D7/FLASH_D7
eMMC_CMD/FLASH_WRn
eMMC_CLKOUT/FLASH_DQS
eMMC_DATA_STROBE/FLASH_CLE
R130 0R 5% R0201
eMMC_DATA_STROBE
TP63 TP_0.7
TP37 TP_0.7
U13
25Q64DWZPIG
WSON_8_6X5_9
FSPI_CS
FSPI_D1
FSPI_D2
FSPI_D3
FSPI_CLK
FSPI_D0
R132 0R 5% R0201
C221 4.7uF X5R 10V C0402
VCCIO_FLASH
R133 10K 5% R0201
R134 10K 5% R0201
J4
CON34A
VCC_3V3
R135 10K 5% R0201
R136 NC R0201 5%
J6
CON2
TP38
FLASH_DTE
ED8 BTR04G02 ESD0402_1xD5
J5
CON2
Maskrom
ED7 BTR04G02 ESD0402_1xD5
R203 2.2K R0201 5%
R204 4.7K R0201 5%
Q10 MMBT3904M SOT723
TP65 T123
R200 2.2K R0201 5%
R201 4.7K R0201 5%
Q9 MMBT3904M SOT723
TP64 T123
C222 100nF X5R 10V C0201
C223 100nF X5R 10V C0201
C224 4.7uF X5R 10V C0402
C225 4.7uF X5R 10V C0402
GND
PINE64
Size A
Title: Quartz64 Model-B Schematic-20220124
Page Name: Flash-eMMC Flash
REV V1.3
Date: Monday, February 21, 2022
Sheet 17 of 26



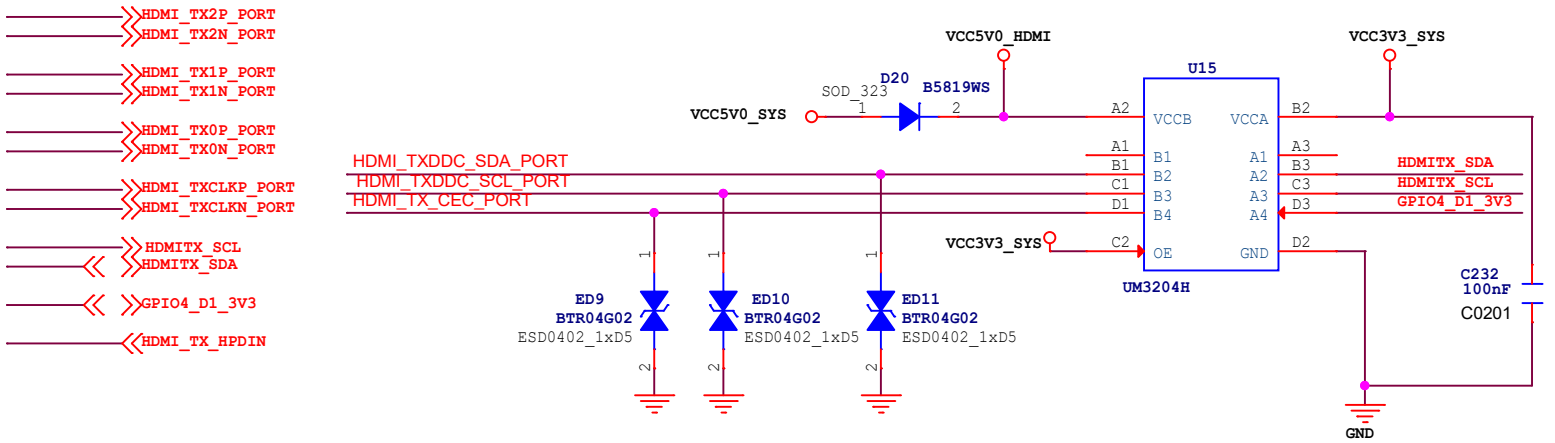
MicroSD Card

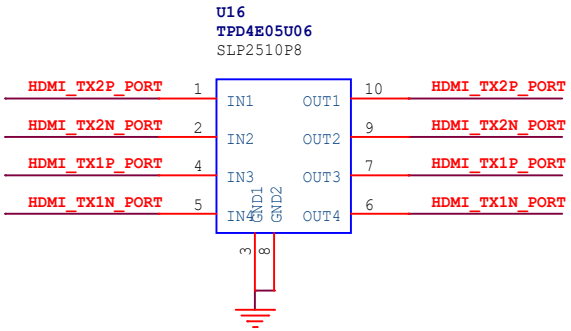
PINE64		
Size: A4	Title: Quartz64 Model-B Schematic-20220124	REV: V1.3
	Page Name: Flash-MicroSD Card	
Date: Monday, January 24, 2022	Sheet 18 of 26	

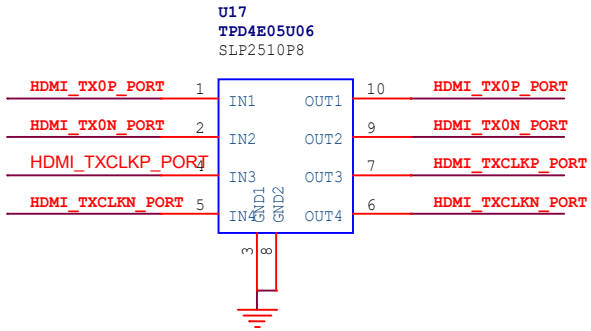


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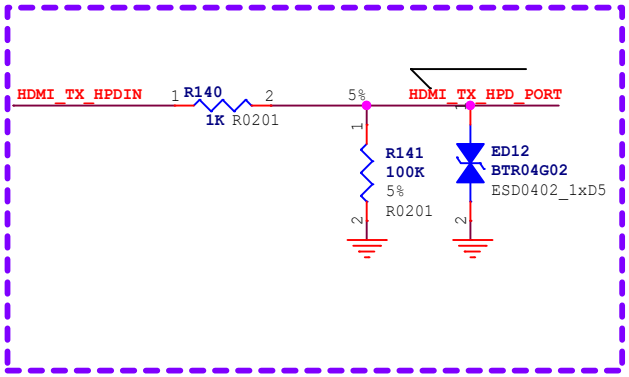
Size: A	Title: Quartz64 Model-B Schematic-20220124	REV: V1.3
	Page Name: VI-Camera_MIPI-CSI	
Date: Monday, January 24, 2022	Sheet 19 of 26	

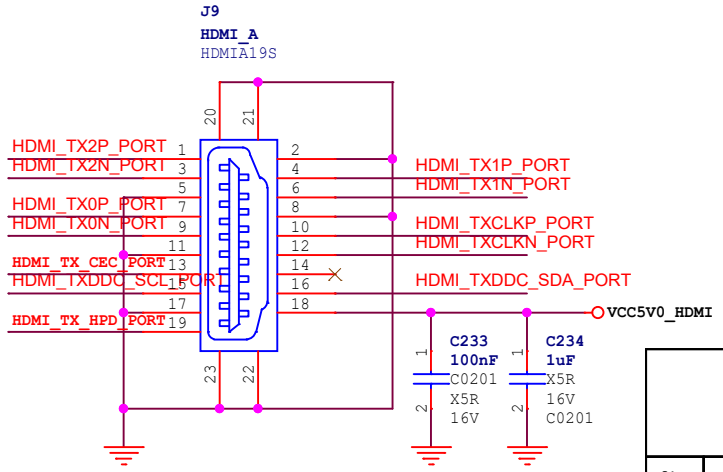






HDMI TX HPD





PINE64

Size: A4	Title: Quartz64 Model-B Schematic-20220124	REV: V1.3
	Page Name: VO-HDMI2.0 TX	
Date: Monday, January 24, 2022	Sheet 20 of 26	

MIPI_DSI_TX1_D0P
MIPI_DSI_TX1_D0N
MIPI_DSI_TX1_D1P
MIPI_DSI_TX1_D1N
MIPI_DSI_TX1_CLKP
MIPI_DSI_TX1_CLKN
LCD0_PWREN_H_GPIO0_C7

CON15

[26] MIPI_DSI_TX1_D1N
[26] MIPI_DSI_TX1_D1P
[26] MIPI_DSI_TX1_CLKN
[26] MIPI_DSI_TX1_CLKP
[26] MIPI_DSI_TX1_D0N
[26] MIPI_DSI_TX1_D0P
[26] I2C3_SCL_M1
[26] I2C3_SDA_M1
VCC_MIPI

J10

VCC_MIPI
C235
10uF
C0603
X5R
10V

VCC3V3_SYS
C236
1uF
X5R 10V
C0201

U18
SY6280AAC
IN
VOUT
GND
EN
OCB
SOT_23_5

[26] LCD0_PWREN_H_GPIO0_C7

R142
4.7K
5%
R0201

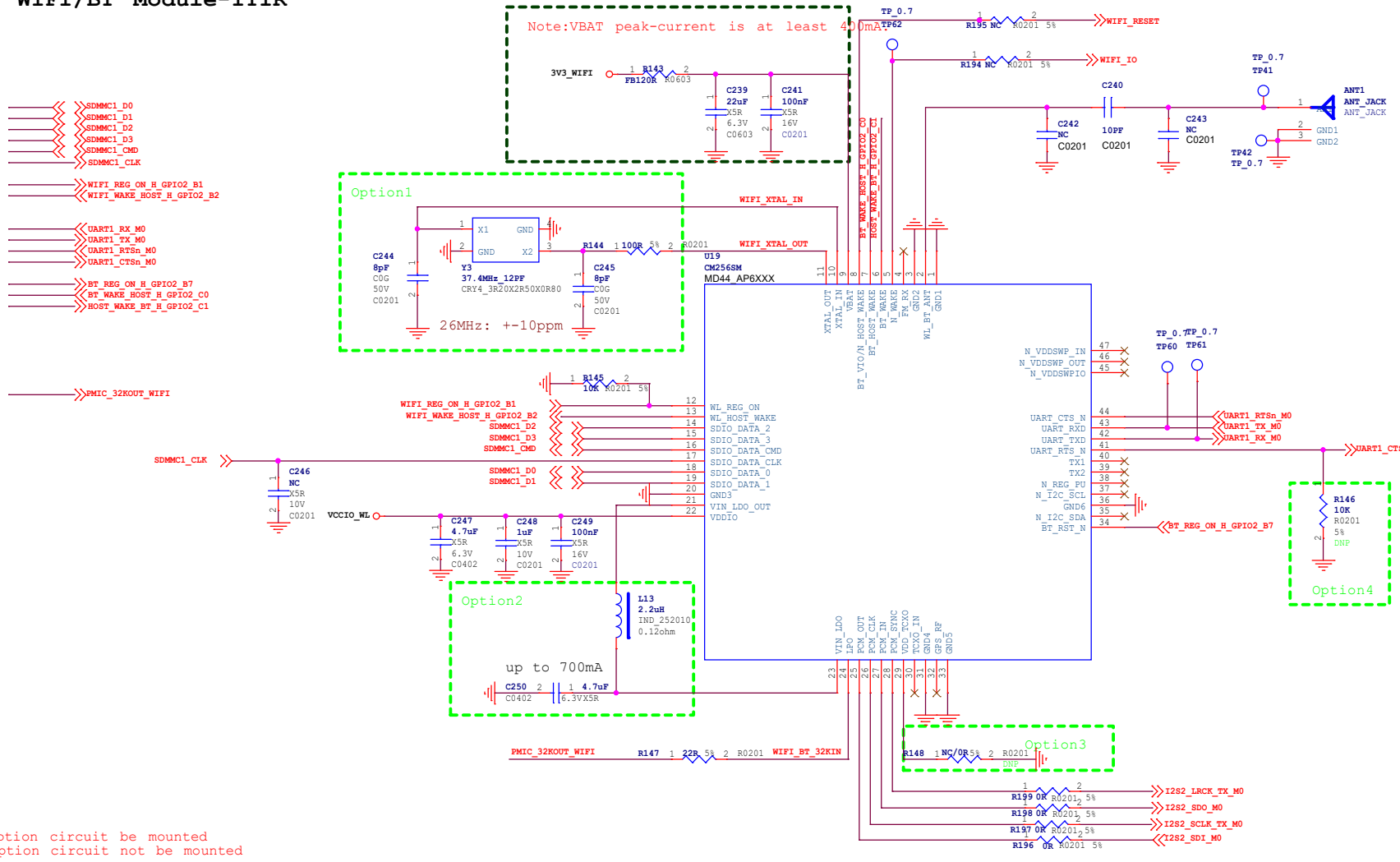
VCC_MIPI
C237
10uF
C0603
10V
X5R

C238
100nF
X5R
10V
C0201

PINE64

Size	Title: Quartz64 Model-B Schematic-20220124	REV
A	Page Name: VO-LCM_MIPI-DSI	V1.3
Date: Monday, January 24, 2022	Sheet 21 of 26	

SDIO WIFI/BT Module-1T1R



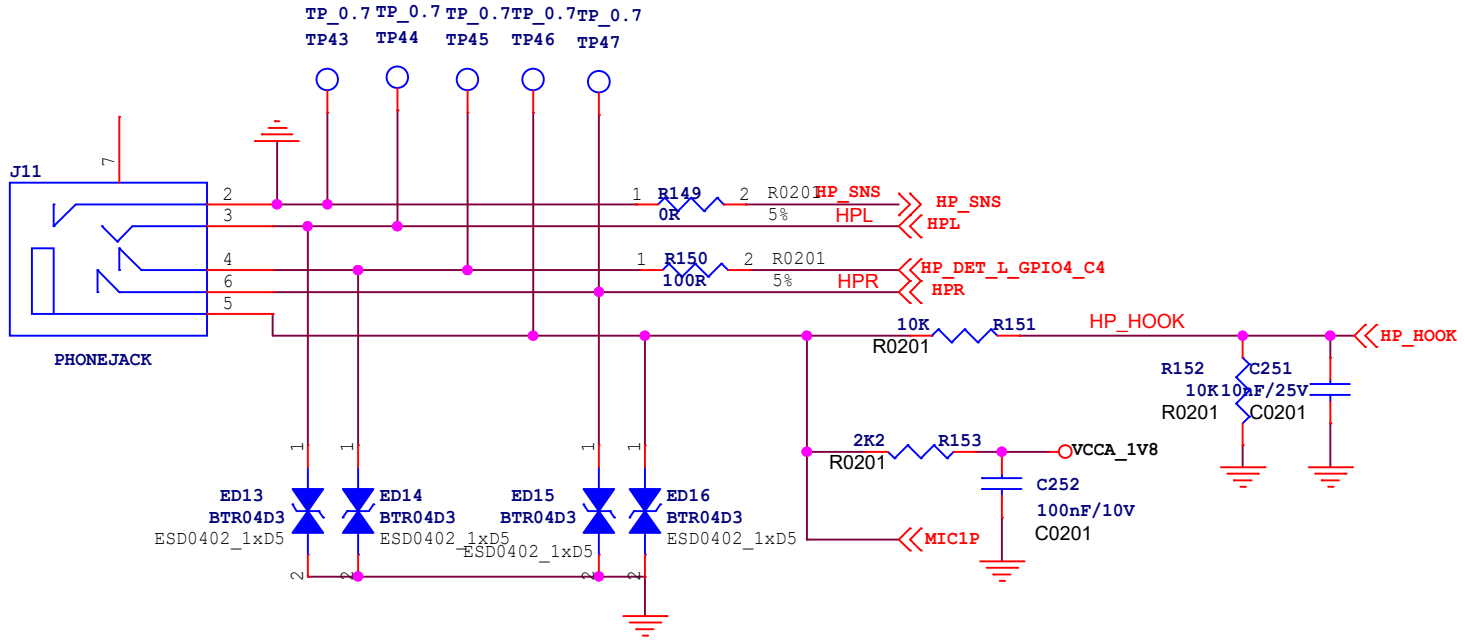
Note:
Yes: option circuit be mounted
No: option circuit not be mounted

OPTION	WIFI				BT	Crystals	VCCIO_SDIO	OPTION1	OPTION2	OPTION3	OPTION4
	a	b/g/n	ac	5GHz							
AW-CM256SM	Yes	Yes	Yes	Yes	4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	No
AP6236/AP6212	No	Yes	No	No	4.2/4.0	26MHz	1.71-3.6V	Yes	Yes	No	No
AP6256/AP6255	Yes	Yes	Yes	Yes	5.0/4.2	37.4MHz	1.71-3.6V	Yes	Yes	Yes@SDIO2.0 No@SDIO3.0	No
RTL8189ETV Module F89FTSM12-W3	No	Yes	No	No	No	Module Integrated	1.8-3.3V	No	No	No	No
RTL8723BS Module F23BDSM23-W2	No	Yes	No	No	4.0	Module Integrated	1.62-3.6V	No	No	No	No
RTL8723DS Module 6223A-SRD	No	Yes	No	No	4.2	Module Integrated	1.62-3.6V	No	No	No	Yes
QCA9377 Module 8223A-SR	Yes	Yes	Yes	Yes	4.2	Module Integrated	1.7-3.45V	No	No	No	No
RTL8821CS Module 6221A-SRC	Yes	Yes	Yes	Yes	4.2	Module Integrated	1.7-3.45V	No	No	No	No

PINE64

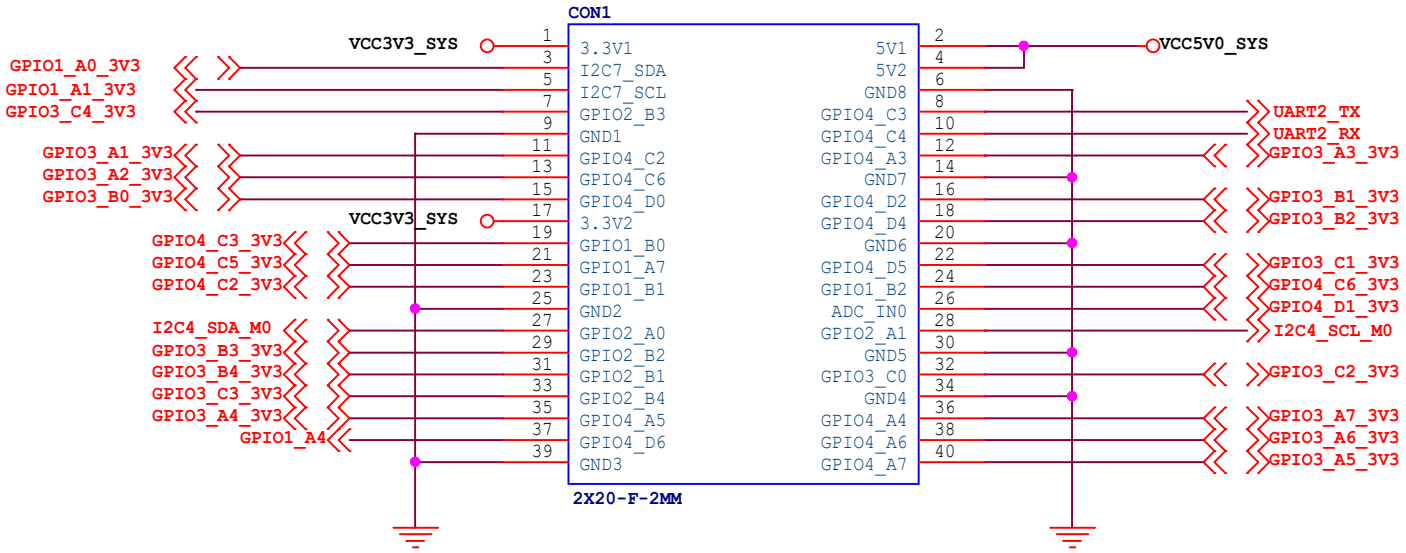
Size	Title: Quartz64 Model-B Schematic-20220124	REV
A3	Page Name: WIFI/BT-SDIO_1T1R	V1.3
Date: Monday, January 24, 2022	Sheet 22 of 26	

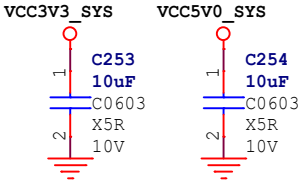
Headphone (Reserved)



PINE64

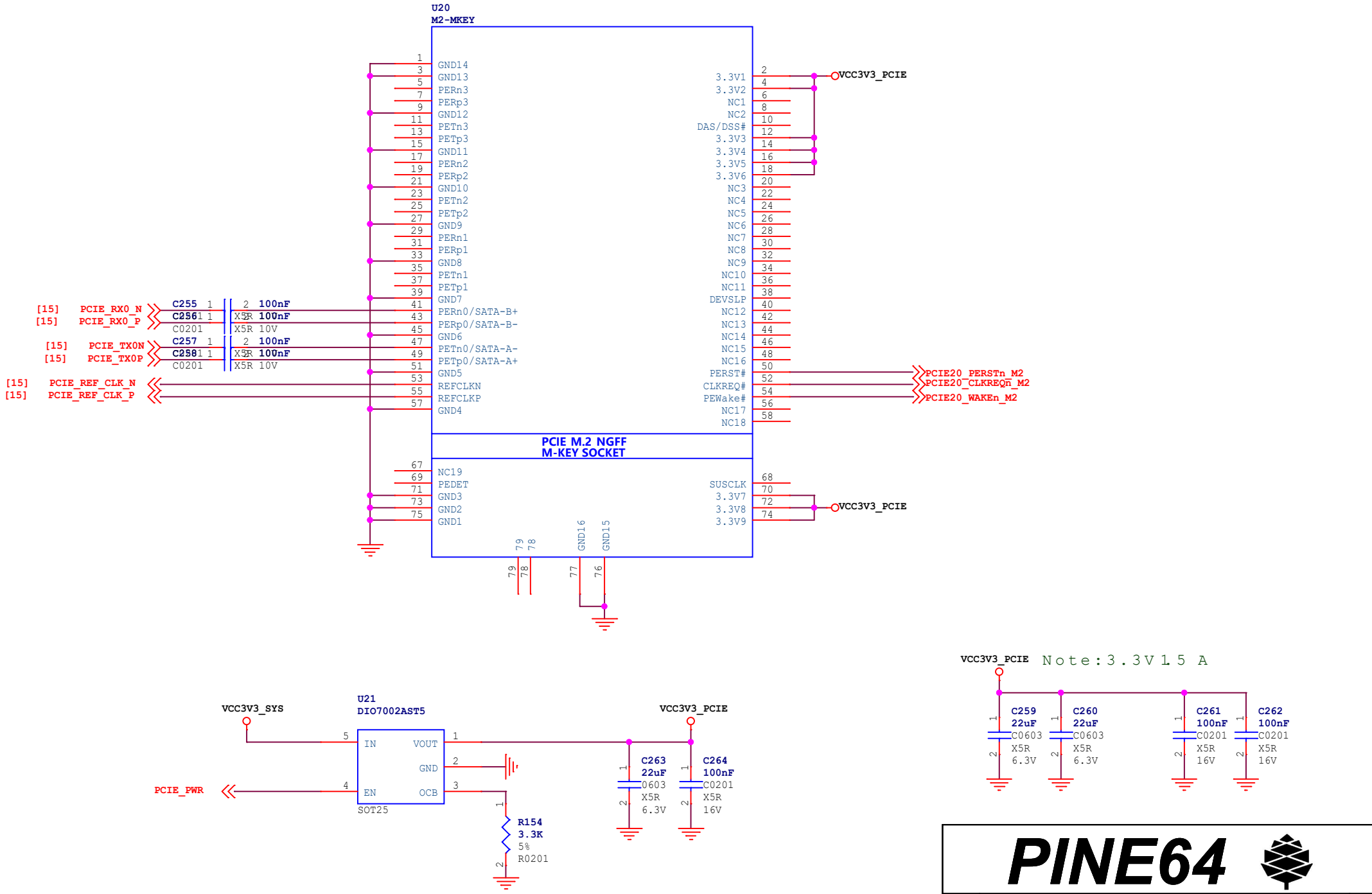
Size	Title: Quartz64 Model-B Schematic-20220124	REV
A	Page Name: AUDIO-HPOUT	V1.3
Date: Monday, January 24, 2022	Sheet 23 of 26	





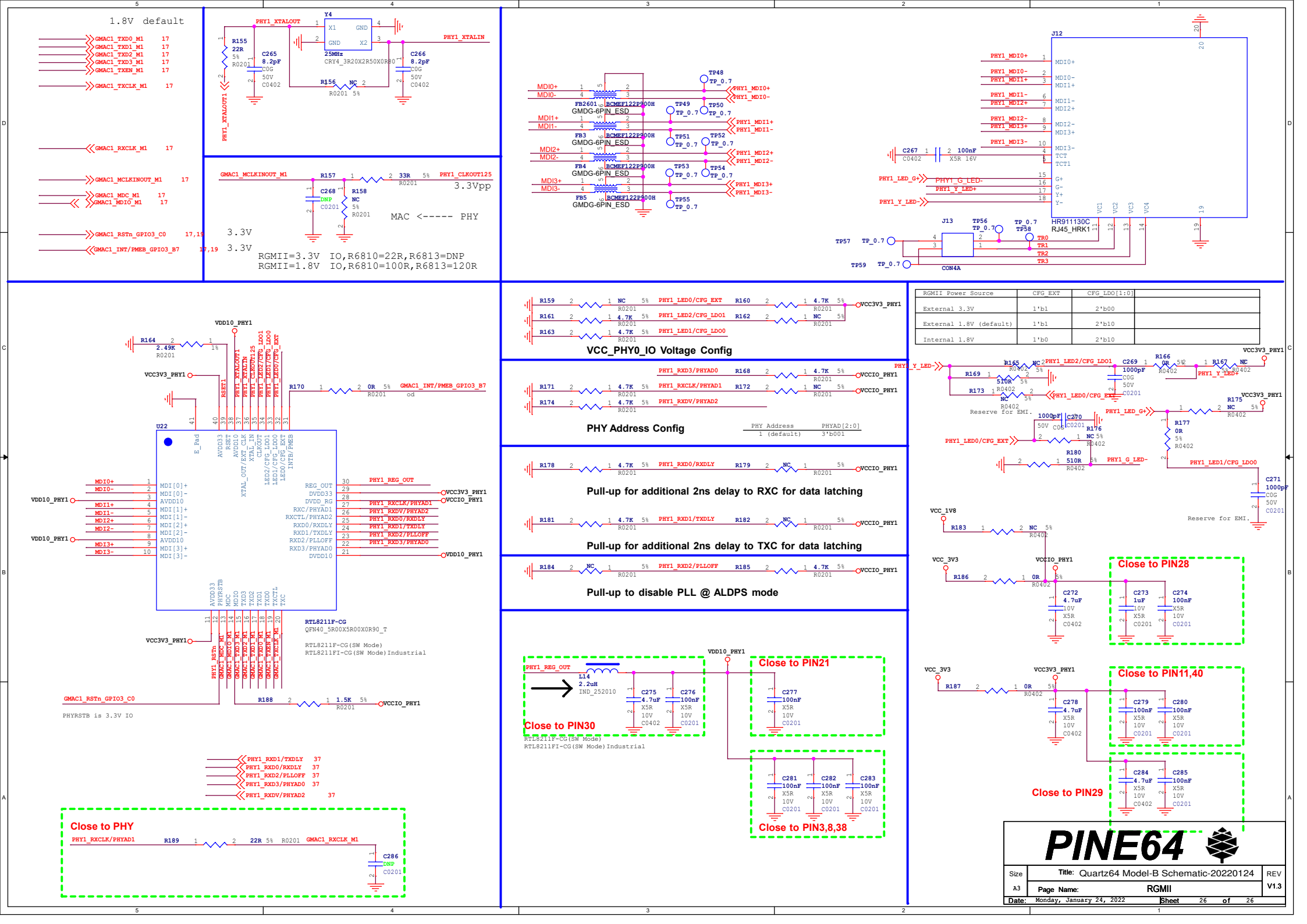
PINE64

Size	Title: Quartz64 Model-B Schematic-20220124	REV
A	Page Name: CONNECTOR	V1.3
Date: Monday, January 24, 2022	Sheet 24 of 26	



PINE64

Size	Title: Quartz64 Model-B Schematic-20220124	REV
A4	Page Name: PCIE-M2	V1.3
Date: Monday, January 24, 2022	Sheet 25 of 26	


1.8V default
GMAC1_TXD0_M1 17
GMAC1_TXD1_M1 17
GMAC1_TXD2_M1 17
GMAC1_TXD3_M1 17
GMAC1_TXEN_M1 17
GMAC1_TXCLK_M1 17
GMAC1_RXCLK_M1 17
GMAC1_MCLKINOUT_M1 17
GMAC1_MDC_M1 17
GMAC1_MDIO_M1 17
GMAC1_RSTn_GPIO3_C0 17,19
GMAC1_INT/PMEB_GPIO3_B7 17,19
3.3V
3.3V
PHY1_XTALOUT
PHY1_XTALIN
Y4
25MHz
CRY4_3R20X2R50X0R80
R155 22R
C265 8.2pF
C266 8.2pF
R156 NC
GMAC1_MCLKINOUT_M1
R157 33R
PHY1_CLKOUT125
3.3Vpp
C268 DNP
R158 NC
MAC <----- PHY
RGMII=3.3V IO,R6810=22R,R6813=DNP
RGMII=1.8V IO,R6810=100R,R6813=120R
MDI0+
MDI0-
MDI1+
MDI1-
MDI2+
MDI2-
MDI3+
MDI3-
FB2601 BCMEF122P900H GMDG-6PIN_ESD
FB3 BCMEF122P900H GMDG-6PIN_ESD
FB4 BCMEF122P900H GMDG-6PIN_ESD
FB5 BCMEF122P900H GMDG-6PIN_ESD
TP48 TP_0.7
TP49 TP_0.7
TP50 TP_0.7
TP51 TP_0.7
TP52 TP_0.7
TP53 TP_0.7
TP54 TP_0.7
TP55 TP_0.7
PHY1_MDI0+
PHY1_MDI0-
PHY1_MDI1+
PHY1_MDI1-
PHY1_MDI2+
PHY1_MDI2-
PHY1_MDI3+
PHY1_MDI3-
J12
HR911130C
RJ45_HRK1
C267 100nF X5R 16V
PHY1_LED_G+
PHY1_G_LED-
PHY1_Y_LED+
PHY1_Y_LED-
J13
CON4A
TP56 TP_0.7
TP57 TP_0.7
TP58 TP_0.7
TP59 TP_0.7
TR0
TR1
TR2
TR3
VCC3V3_PHY1
VDD10_PHY1
R164 2.49K
RSET1
PHY1_XTALOUT
PHY1_XTALIN
PHY1_CLKOUT125
PHY1_LED2/CFG_LDO1
PHY1_LED1/CFG_LDO0
PHY1_LED0/CFG_EXT
R170 0R
GMAC1_INT/PMEB_GPIO3_B7
U22
E_Pad
AVDD33
RSET
AVDD10
XTAL_OUT/EXT_CLK
XTAL_IN
CLKOUT
LED2/CFG_LDO1
LED1/CFG_LDO0
LED0/CFG_EXT
INTB/PMEB
MDI[0]+
MDI[0]-
AVDD10
MDI[1]+
MDI[1]-
MDI[2]+
MDI[2]-
AVDD10
MDI[3]+
MDI[3]-
REG_OUT
DVDD33
DVDD_RG
RXC/PHYAD1
RXCTL/PHYAD2
RXD0/RXDLY
RXD1/TXDLY
RXD2/PLLOFF
RXD3/PHYAD0
DVDD10
PHY1_REG_OUT
PHY1_RXCLK/PHYAD1
PHY1_RXDV/PHYAD2
PHY1_RXD0/RXDLY
PHY1_RXD1/TXDLY
PHY1_RXD2/PLLOFF
PHY1_RXD3/PHYAD0
AVDD33
PHYRSTB
MDC
MDIO
TXD3
TXD2
TXD1
TXD0
TXCTL
TXC
RTL8211F-CG
QFN40_5R00X5R00X0R90_T
RTL8211F-CG(SW Mode)
RTL8211FI-CG(SW Mode)Industrial
PHY1_RSTn
GMAC1_MDC_M1
GMAC1_MDIO_M1
GMAC1_TXD3_M1
GMAC1_TXD2_M1
GMAC1_TXD1_M1
GMAC1_TXD0_M1
GMAC1_TXCTL_M1
GMAC1_TXCLK_M1
GMAC1_RSTn_GPIO3_C0
R188 1.5K
PHYRSTB is 3.3V IO
PHY1_RXD1/TXDLY 37
PHY1_RXD0/RXDLY 37
PHY1_RXD2/PLLOFF 37
PHY1_RXD3/PHYAD0 37
PHY1_RXDV/PHYAD2 37
Close to PHY
PHY1_RXCLK/PHYAD1
R189 22R
GMAC1_RXCLK_M1
C286 DNP
R159 NC
R160 4.7K
R161 4.7K
R162 NC
R163 4.7K
PHY1_LED0/CFG_EXT
PHY1_LED2/CFG_LDO1
PHY1_LED1/CFG_LDO0
VCC3V3_PHY1
VCC_PHY0_IO Voltage Config
R171 4.7K
R172 NC
R174 4.7K
PHY1_RXD3/PHYAD0
PHY1_RXCLK/PHYAD1
PHY1_RXDV/PHYAD2
VCCIO_PHY1
PHY Address Config
PHY Address PHYAD[2:0]
1 (default) 3'b001
R178 4.7K
R179 NC
R181 4.7K
R182 NC
R184 NC
R185 4.7K
PHY1_RXD0/RXDLY
PHY1_RXD1/TXDLY
PHY1_RXD2/PLLOFF
Pull-up for additional 2ns delay to RXC for data latching
Pull-up for additional 2ns delay to TXC for data latching
Pull-up to disable PLL @ ALDPS mode
L14 2.2uH IND_252010
C275 4.7uF
C276 100nF
C277 100nF
C281 100nF
C282 100nF
C283 100nF
Close to PIN30
Close to PIN21
Close to PIN3,8,38
RGMII Power Source | CFG_EXT | CFG_LDO[1:0]
External 3.3V | 1'b1 | 2'b00
External 1.8V (default) | 1'b1 | 2'b10
Internal 1.8V | 1'b0 | 2'b10
R165 NC
R166 0R
R167 NC
R169 510R
R173 NC
C269 1000pF
R175 NC
R176 NC
R177 0R
R180 510R
C270 1000pF
C271 1000pF
PHY1_G_LED-
Reserve for EMI.
VCC_1V8
R183 NC
VCC_3V3
R186 0R
C272 4.7uF
C273 1uF
C274 100nF
Close to PIN28
R187 0R
C278 4.7uF
C279 100nF
C280 100nF
Close to PIN11,40
C284 4.7uF
C285 100nF
Close to PIN29
PINE64
Size A3
Title: Quartz64 Model-B Schematic-20220124
REV V1.3
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